

LOGIC FOR ALL: AN INTRODUCTION TO LOGICAL REASONING



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Logic for All: An Introduction to Logical Reasoning

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Welcome to Logic4All

Your Gateway to Logical Thinking

No Prior Experience Needed

This course is designed for absolute beginners. If you can think, you can learn logic!

Logic4All is an accessible, step-by-step introduction to the fascinating world of logical reasoning. Whether you're a student, professional, or simply curious about improving your thinking skills, this course will equip you with practical tools to analyze arguments, solve problems, and make better decisions.

What is Logic?

Definition

Logic is the systematic study of the principles of valid reasoning and argument. It provides the foundation for distinguishing between good and bad reasoning by establishing rules that govern sound thinking.

Logic is like the grammar of thought. Just as grammar gives structure to language, logic gives structure to reasoning. It helps us organize our ideas, evaluate information critically, and reach reliable conclusions.

Formal Logic

Formal logic deals with the structure of reasoning, using symbols and formulas to represent and analyze arguments.

Example: If all humans are mortal (premise 1), and Socrates is human (premise 2), then Socrates is mortal (conclusion).

Informal Logic

Informal logic focuses on everyday reasoning, analyzing arguments expressed in natural language and identifying common reasoning errors.

Example: Recognizing that "Everyone does it" is not a valid justification for an action (the bandwagon fallacy).

A Brief History of Logic

The development of logic spans thousands of years across multiple civilizations. Below is a timeline of key milestones in logic, from antiquity to the modern era:

```
timeline
  title Evolution of Logic
  4th century BCE: Aristotle's syllogistic logic
  13th century: Scholastic logic in medieval universities
  1854: Boole's Boolean algebra (symbolic logic)
  1879: Frege's predicate logic (first-order logic)
  Early 20th: Russell & Whitehead link logic and mathematics
  1930s: Gödel (incompleteness) and Turing (computing theory)
  1950s: Logic applied to computing and artificial intelligence
  21st century: Logic drives algorithms and AI systems
```

▼ Aristotle's Contribution

Aristotle's systematic approach to logic included:

1. The identification of syllogisms as basic units of reasoning
2. The distinction between inductive and deductive reasoning
3. The formulation of logical fallacies

His work in *Organon* laid the foundation for logical study for over 2,000 years.

Why Study Logic?

Learning logic provides numerous benefits that extend far beyond academic settings:

Practical Benefits

- **Improved critical thinking:** Logic helps you analyze information more effectively and avoid being misled
- **Better decision-making:** Logical thinking leads to more rational choices based on sound reasoning
- **Enhanced problem-solving:** Breaking down complex problems becomes easier with logical methods
- **Stronger communication:** You'll express your ideas more clearly and persuasively
- **Protection against manipulation:** Recognizing faulty reasoning helps defend against deception

Famous Perspective

"Logic is the beginning of wisdom, not the end." — Leonard Nimoy as Spock

Real-life Application

When evaluating claims in advertisements, a logical approach helps you distinguish between factual statements and emotional appeals designed to manipulate your purchasing decisions.

Where Do We Use Logic?

Logic is everywhere in our daily lives and across all fields of knowledge:

1. Everyday Reasoning

We use logic constantly in daily decision-making:

- Deciding what to wear based on weather forecasts
- Planning the most efficient route for running errands
- Budgeting finances and prioritizing expenses
- Evaluating claims on social media for accuracy

Logical Thinking in Action

When your phone stops working, you naturally use logical troubleshooting:

1. Is it out of battery? (If yes, charge it and problem solved)
2. If not, does it turn on at all? (If no, might be a hardware issue)
3. If it turns on but specific functions don't work, which apps or features are affected?

This step-by-step elimination process is logic in action!

2. Academic Disciplines

Logic forms the foundation of many academic fields:

Discipline	How Logic is Used
Mathematics	Proving theorems, solving equations, developing algorithms
Computer Science	Programming, database design, artificial intelligence
Philosophy	Analyzing arguments, examining ethical theories
Law	Building legal arguments, interpreting statutes
Science	Forming hypotheses, designing experiments, analyzing results

3. Professional Applications

Workplace Logic

Logic is essential in:

- **Management:** Strategic planning and resource allocation
- **Engineering:** Designing systems and solving technical problems
- **Medicine:** Diagnosing conditions based on symptoms
- **Finance:** Risk assessment and investment strategies
- **Marketing:** Analyzing consumer behavior and campaign effectiveness

How This Course Works

Logic4All is structured to make learning logic enjoyable and accessible.

Important Note

This course provides an introduction to logic, not a comprehensive study. Our goal is to build a strong foundation that you can expand upon. Along the way, we'll also introduce a few **advanced topics** (like logic with quantifiers and modal logic) and encounter famous **logical paradoxes** to spark your curiosity for further exploration beyond the basics.

Learning Approach

We believe in learning by doing. Each module includes:

- Simple explanations with everyday examples
- Practice exercises to reinforce concepts
- Discussion prompts to deepen understanding
- Knowledge checks to confirm your progress

Learning Outcomes

By the end of this course, you'll be able to:

- Identify the structure of logical arguments
- Evaluate the strength of reasoning in everyday contexts
- Recognize and avoid common logical fallacies
- Apply logical thinking to your own decision-making
- Communicate your reasoning more effectively

Getting Started

Ready to begin your journey into logical thinking? Here's how:

1. **Navigate** using the sidebar to explore topics in order
2. **Engage** with the interactive examples and exercises

3. **Apply** what you learn to situations in your daily life
4. **Discuss** concepts and questions in the community forum

Best Practices

- Study in short, focused sessions
 - Try explaining concepts to someone else
 - Look for examples of logical concepts in daily life
 - Don't rush — build a solid understanding before moving on
-

"The beginning of wisdom is the definition of terms." — Socrates

Let's start our logical journey together!

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Part 1: Logical Reasoning

Why Logical Reasoning Matters

Have you ever tried to figure out why your phone isn't charging, or how to plan a trip so you minimize both time and cost? Chances are you used **logic** without even realizing it. **Logical reasoning** isn't a skill reserved for philosophers and mathematicians; it's something we do every day to make sense of the world around us.

A Simple Metaphor

Think of **logic** like a detective's toolkit. Detectives gather clues, connect the dots, and eliminate impossible scenarios to arrive at the truth. In our daily lives, we often do something similar—just more casually. Formal logic gives us a systematic approach to using this “detective mindset” for stronger, more reliable conclusions.

What Is Deductive Reasoning?

Deductive reasoning is a **top-down** approach: you start with certain statements (premises) and apply rules of reasoning to arrive at a new statement (conclusion). If the premises are true and the reasoning is valid, the conclusion must also be true.

Why It Matters

- **Certainty:** Unlike guessing or intuitive leaps, deductive reasoning helps ensure that if your starting points are correct, you can trust your conclusion.
- **Clarity:** By following a step-by-step process, you avoid ambiguous or contradictory reasoning.
- **Universality:** Deductive methods can be applied to everyday life—like planning, troubleshooting, or decision-making—as well as more formal contexts like mathematics and law.

Deductive Reasoning in Action: The Box of Colored Balls

Let's illustrate how we use deductive reasoning with a simple puzzle. Imagine we have a box with colored balls under these three conditions:

- (1) The box has **at least one ball** in it colored **yellow, red, or blue**.
- (2) If the box has a **red ball**, then it **also** has a **yellow ball** and a **blue ball**.
- (3) If the box has a **blue ball**, then it **also** has a **red ball**.

We want to see what we can deduce from these statements. Specifically, can we conclude any of the following from (1), (2), and (3)?

- (a) The box has a **red** ball.
- (b) The box has a **yellow** ball.
- (c) The box does **not** have a **blue** ball.

An Informal Argument

To determine which statements can be logically derived, let's reason step by step:

1. From **(1)**, we know there is **at least one ball** and that it must be **yellow, red, or blue**.

2. We break it into three cases based on the color of this one known ball:

3. **Case 1: The ball is yellow.**

Then, trivially, the box contains a yellow ball—so (b) is satisfied.

4. **Case 2: The ball is red.**

Then by **(2)**, the box also has a yellow and a blue ball. Therefore, a yellow ball is present—again satisfying (b).

5. **Case 3: The ball is blue.**

Then by **(3)**, the box has a red ball, and by **(2)** (because now we know it has a red ball), it also has a yellow ball. Once again, (b) is satisfied.

In **all** cases, (b) is true. Thus, we can deduce that **the box must have a yellow ball**.

- We **cannot** conclude (a) “the box has a red ball” with certainty because it’s possible the only ball is yellow.
- We **cannot** conclude (c) “the box does not have a blue ball” because some scenarios do include a blue ball.

Conclusion

The only statement we can guarantee is (b): the box has a yellow ball. This is the essence of **deductive reasoning**: when the **structure** of your argument and your **premises** are valid, your conclusion becomes unavoidable.

Informal vs. Formal Reasoning

So far, we’ve used **informal reasoning**—the type of logical thought we might share in everyday conversation or a friendly debate. In mathematics and other precise fields, we use more **formal reasoning**:

- **Informal Logic**: Uses natural language and everyday examples. It’s quick, intuitive, and great for initial understanding or common scenarios.
- **Formal Logic**: Uses a more rigid structure and symbolic language. This allows for clarity and the ability to check correctness step-by-step or even by computer programs.

Why Formal Logic?

- **Precision**: In formal logic, every word and symbol has a strict definition. No ambiguity allowed.
- **Automation**: Because it’s strict, formal logic can be processed by computers (proof-checkers, AI reasoning systems, etc.).
- **Universality**: Concepts in formal logic underlie **mathematics**, **computer science**, **philosophy**, and many other disciplines.

Formal vs. Natural Languages

A **language** consists of a set of words (vocabulary) and rules for combining these words into sentences (grammar). A **formal language** has:

- Well-defined grammar rules that can be checked by a computer.
- Symbols that represent logical connectives like “and,” “or,” “not,” “implies,” etc.

A **natural language** (like English) has a looser structure, which makes it incredibly flexible and rich for communication but can introduce ambiguity into formal proofs.

Common Misconception

*Some people think **formal** means “complicated” or “elitist.” In reality, formal logic is simply **precise**—it uses carefully chosen symbols and rules to avoid misinterpretation.*

From Everyday to the Abstract: Building Logical Structures

When we apply logical reasoning: 1. We focus on **connectives** (like “and,” “or,” “implies”) that shape how statements combine. 2. We check whether a conclusion necessarily follows from given premises, regardless of topic (whether it’s colored balls, numbers, or something else entirely). 3. We keep track of **valid** structures, ensuring that if the premises are true, the conclusion cannot be false.

Still Seems Abstract?

Remember the detective metaphor: once you understand the “grammar” of logic, you can apply it to **any** scenario, from simple puzzles to complex scientific problems.

Common Misconceptions

1. **“If a statement sounds true, it must be logically valid.”**

Reality: A statement can sound correct or be persuasive but still lack proper logical support.

2. **“Formal logic is only for mathematicians.”**

Reality: Anyone can learn formal logic; it’s about clarity of thought, not just numbers.

3. **“You can’t do real-world reasoning in strict symbolic form.”**

Reality: Many real-world scenarios (like legal arguments and software verification) rely on symbolic logic for rigor.

Exercises

Below are some exercises to help you practice informal logical reasoning. Try constructing clear, step-by-step arguments.

1.1. Give an informal argument that a box satisfying conditions (2) and (3), and that has **at least two balls** of different colors (from among blue, red, and yellow), must have at least **three balls**, one of each color.

1.2. Explain why statement (a) (the box has a red ball) **cannot** be derived solely from (1), (2), and (3).

1.3. Explain why statement (c) (the box does not have a blue ball) **cannot** be derived solely from (1), (2), and (3).

1.4. Give an informal argument to show that a box satisfying conditions (2) and (3) which does **not** contain a yellow ball also does **not** contain a blue ball.

1.5. Can you derive “the box has a blue ball” from “the box has a yellow ball,” together with conditions (2) and (3)? Explain your reasoning.

By understanding why logic matters and seeing it in action with an everyday example, you’re already building a solid foundation. In the next sections, we’ll delve deeper into **Propositional Logic**, **Truth Tables**, and other essential tools that will refine your reasoning skills further.

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3: Propositional Logic

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Part 2: Propositional Logic

Why Propositional Logic Matters

Imagine you're organizing a surprise party. You might think to yourself:

- *"If my friend is available (p), then we'll book the venue on Friday (q)."*
- *"But if the weather forecast is bad (r), we may delay or choose a different date."*

Without realizing it, you're using **propositional logic** — piecing together statements (like *"My friend is available"*) and connecting them with **if**, **and**, or **or** to draw practical conclusions.

Why bother formalizing this?

- **Clarity:** By precisely defining statements and connecting them with logical rules, you reduce the chance of confusion and misunderstanding.
- **Universality:** Propositional logic underpins much of mathematics, computer science, and everyday reasoning.
- **Efficiency:** Formalizing your reasoning can help you solve problems more systematically.

Recall from Part 1

We saw how **deductive reasoning** uses premises and logical structure to reach solid conclusions. Now we'll sharpen that method with **propositional logic**, where we assign truth values to statements and see how they combine.

What Is a Proposition?

A **proposition** is a statement that has a definite truth value (either **true** or **false**, but not both). In **propositional logic**, we treat these statements as indivisible units or "building blocks" that we can combine using logical connectives.

Everyday Propositions

- "It is raining right now." (True or false?)
- "My phone battery is at 50%." (True or false?)
- " $2 + 2 = 4$." (True or false?)

Each of these can be true or false at a given time, so each is a proposition.

Non-Propositions

Not every sentence is a proposition. For example: - **Incomplete thought**: " $2 + 2$ " by itself isn't a complete statement with a truth value.

- **Question**: "Is it raining?" isn't true or false — it's a request for information.

- **Command**: "Close the door." is neither true nor false — it's an instruction.

- **Opinion**: "Ice cream is delicious" is subjective; its truth value can differ by person and isn't objectively true or false in the logical sense.

What Counts as a Proposition?

Some people think **any statement** in English is automatically a proposition. But questions, commands, and incomplete sentences do **not** have truth values, so they're **not** propositions.

Connectives: The "Glue" of Logical Statements

Propositional logic features **connectives** — words like **not**, **and**, **or**, **if...then** — that link propositions together to form new, compound statements. The grammar of propositional logic tells us how to place these connectives and how to interpret the resulting statements.

Why Connectives Matter

Just as **grammatical connectors** (like "because," "although," "since") shape complex sentences in English, logical connectives shape complex **logical statements**. They let us build from simple truths ("It is raining") to nuanced claims ("It is raining, and I have an umbrella"), while preserving or transforming truth values in a consistent way.

The Main Logical Connectives

In propositional logic, the fundamental connectives are:

- **Negation** ($\neg$): "not"
- **Conjunction** ($\wedge$): "and"
- **Disjunction** ($\vee$): "or" (in the inclusive sense)
- **Conditional** ($\rightarrow$): "if...then"
- **Biconditional** ($\leftrightarrow$): "if and only if"

In this section, we'll focus on the first three, and cover conditionals and biconditionals in later sections.

The Connective "Not" (Negation)

In propositional logic, "**not**" is represented by the symbol $\neg$. If p is a proposition, then $\neg p$ (read "not p ") has the **opposite** truth value of p .

Rule for Negation:

- If p is **true**, then $\neg p$ is **false**.

- If p is **false**, then $\neg p$ is **true**.

Flipping a Switch

Think of negation like **flipping a light switch**:

- If the light is on (p = true), flipping the switch turns it off ($\neg p$ = false).

- If the light is off (p = false), flipping the switch turns it on ($\neg p$ = true).

Real-Life Example of Negation

- p = "I will go jogging today."
- If this is true, then $\neg p$ = "I will not go jogging today" is false.
- If you end up not going jogging, then p is false and $\neg p$ is true.

Example

Let r be the statement "It is raining." Then $\neg r$ is "It is not raining."

- If " r " ("It is raining") is true, then " $\neg r$ " ("It is not raining") is false.

- If " r " is false, then " $\neg r$ " is true.

Double Negation

If you negate a negation, you return to the original statement. This is called **double negation**:

$$\neg(\neg p) \equiv p$$

For example, "It is not the case that I am not jogging" means the same as "I am jogging."

The Connective "And" (Conjunction)

The **and** connective is symbolized by $\wedge$ (sometimes written as "&"). It asserts that **both** statements it connects are true.

Rule for Conjunction:

- " $p \wedge q$ " is **true** if **both** p and q are true.

- " $p \wedge q$ " is **false** if **either** p is false, q is false, or both are false.

Two Keys to Open a Door

Imagine a locked door that requires **two different keys** to open. If you have **both** keys, the door opens (true). If you're missing **even one** key, the door stays locked (false).

Real-Life Example of Conjunction

- r = "It is raining."
- s = "I have my umbrella."

The conjunction $r \wedge s$ means "It is raining **and** I have my umbrella."

- This new statement is true only if **both** it is raining and you actually have an umbrella with you.

Conjunction in Decision-Making

Conjunction is crucial when all conditions must be met: - "I'll buy this house if it has three bedrooms **and** is close to good schools."

- "The software will run if the installation is complete **and** the system meets the requirements."

- "A student passes the course if their attendance is above 80% **and** they score at least 60% on the final exam."

The Connective "Or" (Disjunction)

The **or** connective is symbolized by $\vee$. In logic (and math), "or" is typically used in the **inclusive** sense:

- " $p \vee q$ " is **true** if p is true, q is true, **or both** are true.
- " $p \vee q$ " is **false** only if **both** p and q are false.

Inclusive vs. Exclusive 'Or'

In everyday English, we often use "or" in an **exclusive** way (e.g., a menu might say "soup or salad?" implying one or the other, but not both). In logic, " $p \vee q$ " does **not** exclude the possibility that **both** p and q are true. If we need to exclude "both," we explicitly specify "*either...or...but not both*" (this is the exclusive or).

Real-Life Example of Disjunction

- r = "It is raining."
- s = "I have my umbrella."

The disjunction $r \vee s$ means "Either it is raining **or** I have my umbrella (or both)."

- This statement is true if it's raining, or if you have your umbrella, or if both are true.

Choosing Activities

Suppose it's the weekend and you think: "I will go to the **movies** or do some **gardening**." In logical terms, this doesn't exclude doing both (maybe you watch a movie in the morning and garden in the afternoon). If you truly want to exclude doing both, you'd say something like "I will do **either** the movies **or** gardening, but **not both**" (an exclusive or, which is a different concept).

Exclusive OR (XOR)

The **exclusive or** (often abbreviated XOR) is a variant of disjunction where both statements cannot be true simultaneously:

- " $p \text{ XOR } q$ " is true if either p is true or q is true, **but not both**.
- In logical notation, we can write XOR as: $((p \vee q) \wedge \neg(p \wedge q))$
- Sometimes the symbol $\oplus$ is used to represent exclusive or.

Parentheses and Precedence

When combining propositions with multiple connectives, we use **parentheses** to clarify grouping. For instance, $p \wedge (q \vee r)$ generally means something different from $(p \wedge q) \vee r$.

Why Precedence Matters

Think of parentheses like clarifying instructions in a recipe: - "Mix (flour and sugar) and then whisk" is different from "(Mix flour) and (sugar and whisk)." The parentheses tell us which actions or ingredients go together first.

Default Precedence Rules: In logic, by convention: - Negation ($\neg$) has **higher precedence** than and ($\wedge$) or or ($\vee$). This means $\neg p \wedge q$ is read as $(\neg p) \wedge q$, not $\neg(p \wedge q)$. Likewise, $p \vee \neg q$ is read as $p \vee (\neg q)$.

- And ($\wedge$) and Or ($\vee$) have higher precedence than $\rightarrow$ and $\leftrightarrow$ (which we'll cover later).

If in doubt, **use parentheses** to make the intended meaning clear.

Removing Unnecessary Parentheses

Starting expression:

```

$$\$(((\neg p) \wedge q) \vee (\neg (r \vee s)))\$$$

```

We can remove some parentheses without changing the meaning: 1. Remove the outermost parentheses:

```

$$\$( (\neg p) \wedge q) \vee (\neg (r \vee s)) \$$$

```

2. Since negation binds tightly, we can rewrite:

```

$$\$(\neg p \wedge q) \vee \neg(r \vee s) \$$$

```

The expression is now simpler but **still clear** in meaning.

Order of Operations in Propositional Logic

Standard precedence (from highest to lowest) is: 1. Parentheses

2. Negation ($\neg$)

3. Conjunction ($\wedge$) and Disjunction ($\vee$)

4. Conditional ($\rightarrow$) and Biconditional ($\leftrightarrow$)

When connectives have the same precedence level (e.g. $\wedge$ and $\vee$), use parentheses to avoid ambiguity.

Applications of Propositional Logic

Propositional logic isn't just abstract — it has practical applications in many fields:

Computer Science

- **Boolean Algebra & Circuits:** Digital circuit design uses the same logical principles (e.g., an AND gate corresponds to $\wedge$, an OR gate to $\vee$).
- **Programming:** Conditional statements (`if - else`) and logical operators (`&&` , `||` , `!` in many languages) mirror propositional logic.
- **Database Queries:** SQL WHERE clauses use logical operators (AND, OR, NOT) to filter data.

Law

- **Legal Reasoning:** Breaking down complex laws into combinations of conditions (propositions) with logical structure (e.g., "If act X and circumstance Y, then consequence Z").
- **Contract Analysis:** Ensuring contracts spell out conditions and outcomes clearly using logical structure ("if and only if" appears often to set exact conditions).

Mathematics

- **Set Theory & Counting:** Logical connectors relate to set operations (AND $\sim$ intersection, OR $\sim$ union).
- **Proof Techniques:** Constructing valid mathematical proofs often boils down to using propositional logic correctly (e.g., proof by contradiction uses negation).

Everyday Reasoning

- **Decision Trees:** We often draw simple flowcharts for decisions that mimic logical statements (if condition, then outcome).
- **Critical Thinking:** Evaluating arguments in debates or articles involves checking if the conclusion really follows from the premises (propositional logic helps reveal hidden assumptions or fallacies).

Common Misconceptions

- **"If p is false, then $p \rightarrow q$ must be false."**
Reality: If p is false, the statement $p \rightarrow q$ is considered **true** (because a false hypothesis trivially makes the implication true – see the truth table in Part 4).
- **"In logic, 'or' means only one or the other, not both."**
Reality: Unless explicitly stated as exclusive, "or" in logic is inclusive. $p \vee q$ allows for both p and q being true.
- **"Formalizing statements is unnecessary work."**
Reality: Writing statements symbolically can reveal ambiguities or hidden assumptions that are easy to overlook in natural language.

Exercises

Understanding Connectives

For each informal statement, identify the logical connectives and rewrite it using logical symbols (let p , q , r represent basic propositions):

2.1. "I will go to the park if it is sunny and I have finished my homework."

2.2. "Either we win this game or we learn a valuable lesson (or both)."

2.3. "It is not true that both team A and team B can win the championship."

Translating to English

Describe in plain English what the following propositional logic statements mean:

2.4. $\neg p \vee q$

2.5. $(p \wedge q) \rightarrow \neg r$

2.6. $p \leftrightarrow q$

Applications

2.7. Give a real-world scenario that could be modeled by the expression $(p \wedge q) \vee r$. Explain what p , q , and r represent in your scenario.

2.8. Write a simple set of conditions (like a mini-puzzle or rule set) and express them with propositional logic. For example, "If the light is on and the switch works, then electricity is flowing."

Next: In **Part 3: Truth Tables**, we'll learn a systematic method to determine the truth of complex statements under every possible scenario.

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4: Truth Tables

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Part 3: Truth Tables

Why Truth Tables Matter

Remember playing a board game where you consider all possible moves before deciding? **Truth tables** do something similar for logical statements. They list every possible combination of truth values for your basic propositions (like "It's raining" or "I have an umbrella") and show whether a larger, compound statement (like "It's raining AND I have an umbrella") is true or false in each scenario.

A Grid of Possibilities

Think of a truth table like a **map of all scenarios**:

If you consider every possible weather condition (raining or not) and whether you have an umbrella or not, a truth table lets you check every combination to see when "I get wet" or "I stay dry" would be true. This grid ensures you leave **no** combination unchecked.

Building on What We've Learned

In **Part 2**, we introduced three connectives: - **Negation**: "not" ($\neg$)

- **Conjunction**: "and" ($\wedge$)

- **Disjunction**: "or" ($\vee$, used inclusively)

Now, we'll **systematically** show how each connective behaves in every possible scenario of truth and falsehood. This methodical approach helps us avoid mistakes (like confusing exclusive "or" with inclusive "or").

Connecting Back

From **Part 2**, recall:

- $\neg p$ flips p 's truth value.

- $p \wedge q$ is true only if **both** p and q are true.

- $p \vee q$ is true if **either** p or q (or both) are true.

Truth tables will make these rules crystal clear by enumerating all possibilities.

Truth Table for Negation ($\neg$)

Consider a single proposition p . It can be either **true (T)** or **false (F)**. A simple table for p and its negation $\neg p$ is:

p	$\neg p$
-----	----------

p	$\neg p$
T	F
F	T

Reading the rows: 1. If p is **true**, then $\neg p$ is **false**.

2. If p is **false**, then $\neg p$ is **true**.

Flipping a Coin

- Let p = "The coin shows heads."
- Then $\neg p$ = "It is not the case that the coin shows heads" (in other words, "The coin shows tails").

If p is true (the coin is heads), $\neg p$ is false (not heads means tails, which is false in this scenario). If p is false (the coin is not heads, i.e., tails), then $\neg p$ is true.

Truth Table for Conjunction ($p \wedge q$)

When we deal with **two propositions** p and q , each can be either T or F. This means there are **four** possible combinations of truth values. The truth table for $p \wedge q$ is:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Interpretation: $p \wedge q$ is true only in the first row, where **both** p and q are true. In every other case, at least one of p or q is false, making $p \wedge q$ false.

Two Conditions for a Concert

Suppose:

- p = "We have tickets."
- q = "The venue is open."

Then $p \wedge q$ = "We have tickets **and** the venue is open." For the concert outing to be successful (true), **both** conditions must hold:

- If you don't have tickets, you can't get in (even if the venue is open).
- If the venue is closed, you can't get in even if you have tickets.

The table shows that only when both p and q are T (tickets and open) is $p \wedge q$ true.

Truth Table for Disjunction ($p \vee q$)

Again, consider two propositions p and q . The truth table for the inclusive "or" ($\vee$) is:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Interpretation: $p \vee q$ is **false** only in the last row (when *both* p and q are false). In all other cases (at least one true), $p \vee q$ is true.

Inclusive vs. Exclusive Recap

Remember, this table is for inclusive "or". If we wanted an exclusive or (true when exactly one of p or q is true, but not both), the truth values in the first row would be different. (Exclusive or would give F when both are true.) Inclusive "or" covers the case where both can be true.

Example:

- p = "I have a discount coupon."

- q = "I have enough cash."

$p \vee q$ = "Either I have a discount coupon **or** I have enough cash (or both)."

- If I have at least one of those, the statement is true. Only if I have neither (no coupon and not enough cash) is the statement false.

Compound Truth Tables

We can create truth tables for more complex expressions that combine multiple connectives. For example, let's analyze $\neg p \vee (p \wedge q)$.

Example: $\neg p \vee (p \wedge q)$

To build the table, list all combinations for p and q , then compute sub-expressions:

p	q	$\neg p$	$p \wedge q$	$\neg p \vee (p \wedge q)$
T	T	F	T	T
T	F	F	F	F
F	T	T	F	T
F	F	T	F	T

How the table is constructed: 1. List all possible truth combinations for p and q (the first two columns).

2. Compute $\neg p$ for each row.

3. Compute $p \wedge q$ for each row.

4. Finally, determine $\neg p \vee (p \wedge q)$ using the results from the previous two columns.

Notice in the final column, the result is T for every row except one. In fact, $\neg p \vee (p \wedge q)$ is true in all cases **except** when p is true and q is false (that was the second row). If you look closely, the final column is all T's except one F.

In the final column, being *all* T in every row would mean the expression is a tautology (always true). Here it's not all T, but mostly. We won't draw conclusions yet, but keep this in mind for later when we discuss special types of compound statements.

This methodical approach (breaking down complex statements into parts and evaluating each combination) helps us **unravel complex logical expressions** without guesswork.

Truth Tables in Decision-Making and Analysis

Truth tables have practical uses beyond homework exercises. They ensure no scenario is overlooked:

Validating Arguments

By using a truth table, you can check whether an argument's conclusion follows from its premises. If there's **no** row where all premises are true and the conclusion is false, the argument is logically valid.

Circuit Design

In digital electronics, truth tables describe how logic gates (AND, OR, NOT, etc.) operate. Engineers design and verify circuits (like computer processors) using truth tables to ensure the outputs are correct for all input combinations.

Software Testing

Programmers use truth-table-like thinking to test all combinations of certain conditions in code (especially complex `if` conditions) to ensure a program behaves correctly in every case. This is akin to covering every row of a truth table.

Legal Analysis

To analyze complex laws with multiple conditions, legal experts might use a structure like a truth table to map out when certain legal outcomes apply. This helps clarify how different facts of a case influence the verdict.

Software Testing Application

Consider a login system with two checks:

- p : "User entered the correct password"
- q : "User passed two-factor authentication (2FA)"

Suppose the access rule is: "A user is allowed in if **(they entered the correct password AND passed 2FA) OR (they entered the correct password AND have an admin override)**." In logical form, if we let r = "User has admin override", the rule could be $(p \wedge q) \vee (p \wedge r)$.

A truth table for $(p \wedge q) \vee (p \wedge r)$ helps testers ensure all scenarios (both conditions met, one met, none met, etc.) yield the expected outcome (allow or deny access).

Common Misconceptions

- **"The logical 'or' must exclude both being true."**
Reality: By default, logical "or" is inclusive; both can be true. Exclusive or is a special case we have to specify separately.
- **"Truth tables are only for math classes, not real life."**
Reality: Truth tables (or the systematic thinking behind them) clarify any situation with clear yes/no conditions, from planning events to troubleshooting electronics.
- **"If there are many propositions, truth tables are impossible to use."**
Reality: It's true that truth tables grow quickly (2^n rows for n propositions). For large n , other techniques or software are used. But for small numbers of propositions, truth tables are a straightforward and reliable tool.
- **"A truth table can be done arbitrarily; there's no standard method."**
Reality: There is a standard method: list all combinations of truth values for the basic propositions, then compute stepwise. This ensures consistency and that no case is missed.

Exercises

- 3.1. Construct a truth table for **exclusive or** (XOR) of two propositions p and q . *Hint:* XOR is true when exactly one of p or q is true (and false otherwise). Verify that your final column matches the definition (true for one true input, false for both true or both false).
- 3.2. Build a truth table for $(p \wedge \neg q) \rightarrow \neg r$. How many rows will it have? (List them all.) Which rows make the implication false?
- 3.3. Using a truth table, determine if the expression $p \vee (q \wedge \neg q)$ is a tautology, contradiction, or contingency (see Part 8 for definitions of these terms, if needed).
- 3.4. Write the conditions for a simple household alarm system and then represent them in a truth table. For example: " a = motion sensor detects movement, b = door sensor detects door open. Alarm sounds if $a \vee b$."
- 3.5. (Challenge) If a compound statement has 3 distinct propositions, how many rows will a complete truth table have? If it has 4 distinct propositions? Explain the pattern.

Next: In **Part 4: If-Then Statements**, we'll focus on the conditional connective ($\rightarrow$), learn its truth table, and see why it's defined the way it is.

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5: If-Then Statements

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Part 4: If-Then Statements

Why If-Then Statements Matter

Imagine you have a **membership card** to an exclusive club. The rule says:

┆ *"If you have the membership card, **then** you can enter the club."*

This is an **if-then** statement (a **conditional** or **implication**). It connects having a membership card (**hypothesis**) to being allowed inside (**conclusion**). If-then statements are everywhere in daily life and mathematics. They help us set **conditions** and figure out **consequences** — essential for problem-solving and structured thinking.

Lock and Key

Think of **if-then** as a **lock-and-key** system:

The **"if"** part (hypothesis) is the **key** you might have.

The **"then"** part (conclusion) is the **lock** that opens (true) or stays closed (false) depending on whether you have the right key.

What Is an If-Then Statement?

An **if-then** statement has two parts: 1. **Hypothesis (if-part)**: The condition that might be true or false.

2. **Conclusion (then-part)**: What is claimed to follow if the hypothesis holds.

In symbolic form, "**if** p **then** q " is written as $p \rightarrow q$, where: - p = the hypothesis (antecedent)

- q = the conclusion (consequent)

Everyday Language Examples

- **"If you jump off the cliff, then you will get hurt."**
Hypothesis: "You jump off the cliff."
Conclusion: "You will get hurt."
- **"What goes up comes down."**
(Can be phrased as: "If something goes up, then it comes down.")
- **"Bigger engines burn more gas."**
(As an implication: "If an engine is bigger, then it burns more gas.")

Other Ways to Say It

" $p \rightarrow q$ " can be read as:

- " p **implies** q ."
- "Whenever p , q follows."
- " p being true **forces** q to be true."

Truth of If-Then: It's Not About Causality

In everyday life, we often expect an **if-then** to reflect a cause-and-effect relationship (if you water plants, they grow). However, in **formal logic**, the truth of $p \rightarrow q$ depends purely on the truth values of p and q , not on actual causation or relevance. By definition, $p \rightarrow q$ is false **only** in one scenario: when p is **true** and q is **false**. In every other case, $p \rightarrow q$ is true.

Consider these four examples of " $p \rightarrow q$ ":

- **If the sky is blue, then the oceans are wet.**
 p = "The sky is blue" (true), q = "The oceans are wet" (true).
 True hypothesis, true conclusion $\rightarrow$ the if-then statement is **true** (even though one doesn't *cause* the other).
- **If the sky is neon yellow, then the oceans are wet.**
 p = "The sky is neon yellow" (false), q = "The oceans are wet" (true).
 Hypothesis is false, so $p \rightarrow q$ is considered **true** (a false "if" makes the implication true, regardless of q).
- **If the sky is neon yellow, then the oceans are dry.**
 p = "The sky is neon yellow" (false), q = "The oceans are dry" (false).
 Hypothesis is false, q is false, but still $p \rightarrow q$ is **true** (because a false p makes $p \rightarrow q$ true, regardless of q).
- **If the sky is blue, then the oceans are dry.**
 p = "The sky is blue" (true), q = "The oceans are dry" (false).
 Here, hypothesis is true and conclusion is false $\rightarrow p \rightarrow q$ is **false**.

Key Truth Rule for Implications

$p \rightarrow q$ is **false only** when p is **true** and q is **false**. In all other cases, $p \rightarrow q$ is true.

Truth Table for If-Then

To confirm the rule, let's see the truth table for a conditional $p \rightarrow q$:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Row by row: - **Row 1:** $p = T, q = T \rightarrow p \rightarrow q = T$. (True implies true is true.)

- **Row 2:** $p = T, q = F \rightarrow p \rightarrow q = F$. (True implies false is the one false case.)

- **Row 3:** $p = F, q = T \rightarrow p \rightarrow q = T$ (if-part false makes the implication true, regardless of q).

- **Row 4:** $p = F, q = F \rightarrow p \rightarrow q = T$ (hypothesis false again makes the implication true).

Far-Fetched Hypothesis

Let p = "Dogs can read" (false), and q = "Cats can sing" (false).

The implication $p \rightarrow q$ is "If dogs can read, then cats can sing." According to the truth table, since p is false, $p \rightarrow q$ is **true** in formal logic (even though the content is fanciful). This illustrates that logical truth is about structure, not actual real-world causation.

Using If-Then in Reasoning

In logic, if we know $p \rightarrow q$ is true *and* we know p is true, then we can **deduce** q . This form of reasoning is called **Modus Ponens** — one of the most fundamental inference rules:

From " $p \rightarrow q$ " and " p ," infer " q ."

Real-Life Example (Modus Ponens)

- p = "I have a valid ticket."
- q = "I can enter the concert."

If "Having a valid ticket implies I can enter the concert" ($p \rightarrow q$ is true), and indeed I **do** have a valid ticket (p is true), then I can conclude I am allowed in (q is true).

Another common reasoning pattern is the **contrapositive** form (covered in Part 5): from $p \rightarrow q$ and $\neg q$, infer $\neg p$ (this is **Modus Tollens**). For example, "If I have a membership card, then I can enter. I could not enter, therefore I must not have had a membership card."

Common Misconceptions

- "If p is false, then $p \rightarrow q$ must be false."
Reality: If p is false, $p \rightarrow q$ is **true** by the logical definition (no matter what q is).
- "An if-then statement must reflect a real cause-and-effect."
Reality: Formal logic cares only about truth values, not actual causality or relevance. Even a nonsensical implication can be true in the logical sense if the condition is false or the conclusion is true.
- "If $p \rightarrow q$ is true, then p causes q ."
Reality: " $p \rightarrow q$ " simply means "if p is true, then q is true." It says nothing about causation. (For example, "If $2+2=5$, then pigs can fly" is logically true because $2+2=5$ is false, but there's no causal link intended.)

Historical Context

The concept of conditional statements dates back to ancient Greek logic. Aristotle was among the first to formalize logical reasoning, and the Stoic philosophers further developed propositional logic including early ideas of the if-then statement.

In modern symbolic logic, the implication operator ($\rightarrow$) was standardized in the late 19th and early 20th centuries by logicians like Gottlob Frege, Bertrand Russell, and Alfred North Whitehead as part of their work to formalize mathematics. This eventually fed into computer science (where if-then rules are fundamental to programming).

Applications in Different Fields

Mathematics: Mathematical theorems are often stated as conditionals ("If a number is prime and greater than 2, then it is odd"). Mastering if-then helps in understanding proofs and the concept of necessary/sufficient conditions.

Computer Science: Conditional statements (`if - then - else`) are the backbone of decision-making in algorithms and programs. Knowing logical implications helps programmers avoid bugs in complex conditions.

Law: Many laws are conditional ("If a person commits act X, then consequence Y follows"). Understanding the logic helps in interpreting statutes and constructing legal arguments without logical loopholes.

Medicine: Diagnostic reasoning uses implications ("If the patient shows symptoms X, Y, and Z, then they have condition A"). Recognizing these as if-then rules clarifies why certain tests or treatments are applied.

Exercises

Converting to If-Then Form

Rewrite each statement in a clear *if-then* form ($p \rightarrow q$). Identify the hypothesis and conclusion in each case:

- 4.1. "By getting enough votes you will win the election."
- 4.2. "On the fourth Thursday in November we eat turkey."
- 4.3. "I am happy when I am studying calculus."
- 4.4. "People taller than six feet weigh more than three pounds."
- 4.5. "I will attend school in the fall if I am accepted at Stanford."
- 4.6. "A valid ID is required for admission." (Hint: think of it as "If a person does not have a valid ID, then...")
- 4.7. "I must pay my rent or I will be evicted." (Hint: rewrite as an implication by contrapositive: "If I don't pay rent, then...")

Identifying Hypotheses and Conclusions

For each implication below, label the **hypothesis** (if-part) and the **conclusion** (then-part):

- 4.8. If it rains today, the picnic will be canceled.
- 4.9. If Tom is a bachelor, then Tom is unmarried.
- 4.10. If the alarm is ringing, then a door or window was opened.
- 4.11. If Alice passes the exam, then Bob will throw a party.

Reasoning with Implications

- 4.12. *Modus Ponens*: Given "If I finish my work (p), then I will go out (q)" and "I finished my work (p)," what can you conclude?
- 4.13. *Modus Tollens*: Given "If the cake is made with sugar (p), then it will ferment (q)" and "The cake did not ferment ($\neg q$)," what can you conclude about p ?
- 4.14. Explain in your own words why an implication with a false hypothesis is considered true (use an example different from ones given above).

Next: In Part 5: The Converse and the Contrapositive, we will explore what happens when we flip or negate if-then statements — and why some transformations preserve truth while others do not.

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6: The Converse and the Contrapositive

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Part 5: The Converse and the Contrapositive

Why Converses and Contrapositives Matter

Now that we understand *if-then* statements, let's explore two important variations:

- **The Converse:** Reversing the hypothesis and conclusion.
- **The Contrapositive:** Both reversing **and** negating the hypothesis and conclusion.

Rewinding a Scene

Think of an if-then statement like a scene in a movie:

Converse: You play the scene **backwards** — switching the roles of condition and result. Often, it tells a very **different story**.

Contrapositive: You not only rewind the scene but also **flip** the roles of the characters to their opposites. Surprisingly, this tends to tell the **same truth** as the original scene, just from a different angle.

The Converse: $q \rightarrow p$ (instead of $p \rightarrow q$)

Given an original statement:

$$p \rightarrow q,$$

the **converse** is:

$$q \rightarrow p.$$

In words: - Original: "If p , then q ."

- Converse: "If q , then p ."

Example:

Original: "If it's a goat (p), then it's an animal (q)."

Converse: "If it's an animal (q), then it's a goat (p)."

Does the Converse Have the Same Truth?

The truth of " $p \rightarrow q$ " does **not** guarantee the truth of " $q \rightarrow p$." They are separate statements.

Goat vs. Animal

Original ($p \rightarrow q$): "If it's a goat, then it's an animal." This is **true** (all goats are animals).
 Converse ($q \rightarrow p$): "If it's an animal, then it's a goat." This is **false** — there are many animals that are **not** goats (dogs, cats, etc.).

Key Point:

Knowing that $p \rightarrow q$ is true tells us nothing for sure about $q \rightarrow p$. The converse might be true or false independently.

Truth Table Perspective

Let's compare $p \rightarrow q$ and $q \rightarrow p$ side by side:

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Notice the middle two lines: - **Line 2:** $p \rightarrow q$ is false ($T \rightarrow F$ is false), but $q \rightarrow p$ is true ($F \rightarrow T$ is true).

- **Line 3:** $p \rightarrow q$ is true ($F \rightarrow T$ is true), but $q \rightarrow p$ is false ($T \rightarrow F$ is false).

These mismatches show that $p \rightarrow q$ and $q \rightarrow p$ can differ in truth value.

The Contrapositive: $\neg q \rightarrow \neg p$

The **contrapositive** of:

$$p \rightarrow q$$

is:

$$\neg q \rightarrow \neg p.$$

In words, we take the original if-then ("If p , then q "), and form "If **not** q , then **not** p ."

Surprisingly, a **statement and its contrapositive are logically equivalent** — they are always true or false together.

Box of Balls Revisited

Original: "If the box has a **blue** ball (p), then it has a **red** ball (q)."

Contrapositive: "If the box **does not** have a red ball ($\neg q$), then it **does not** have a blue ball ($\neg p$)."

If the original statement about the box is true, its contrapositive is also true (and vice versa). They stand or fall together.

Why Are They Equivalent?

Intuitive Explanation

If " $p \rightarrow q$ " fails, it means p happened but q did not (that's the only failure case). But in that exact scenario, " $\neg q \rightarrow \neg p$ " also fails (because $\neg q$ is true but $\neg p$ is false). Whenever one fails, the other fails in the same case. In all other cases, both hold true.

Truth Table Confirmation

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
F	F	T	T	T	T

Compare the columns for $p \rightarrow q$ and $\neg q \rightarrow \neg p$ — they are identical in every row (T, F, T, T). That means $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are **logically equivalent** statements.

Equivalence in Practice

In proofs (and many real-life arguments), it's often useful to prove $p \rightarrow q$ by proving $\neg q \rightarrow \neg p$ instead. Sometimes it's easier to assume the outcome doesn't happen (assume q is false) and show the condition couldn't have happened (p must be false).

The Inverse: $\neg p \rightarrow \neg q$

For completeness, consider the **inverse** of the original statement $p \rightarrow q$, which is:

$$\neg p \rightarrow \neg q.$$

The inverse says: "If **not** p , then **not** q ." The inverse is actually the converse of the contrapositive (or equivalently, the negation of both parts of the converse). The truth of the inverse is **not** guaranteed by the original statement (much like the converse).

Example: Original: "If it's a goat, then it's an animal." (True)

Inverse: "If it's not a goat, then it's not an animal." (False, since not-goat could be a cow, which is still an animal.)

In general, a statement and its inverse are not equivalent. (In fact, the inverse has the same truth value as the converse in all cases.)

Common Misconceptions

- **"The converse has the same meaning as the original."**

Reality: They can differ drastically. $p \rightarrow q$ and $q \rightarrow p$ are usually not equivalent (unless p and q happen to be logically equivalent statements themselves).

- **"If $p \rightarrow q$ is true, then $q \rightarrow p$ must be true."**

Reality: False. The goat/animal example shows $p \rightarrow q$ can be true while $q \rightarrow p$ is false.

- **"The contrapositive is basically the same as the converse."**

Reality: Not at all. The contrapositive of $p \rightarrow q$ is logically **equivalent** to the original statement, whereas the converse is generally not. It's a common error to mix these up.

Exercises

Below are exercises to practice identifying converses and contrapositives, and to test your understanding of their truth values.

Identify Converse and Contrapositive

5.1. For the statement "If you are wearing a red hat, then the people next to you are smiling," write the **converse** and the **contrapositive**.

5.2. Given "If a number is even, then it is divisible by 2," what are the converse and contrapositive? Which of those are true statements in general mathematics?

5.3. Consider the statement "If it is raining, then the ground is wet." Write its converse, inverse, and contrapositive in English sentences.

Truth Check

Determine whether each of the following is true or false (in general). Assume a universe where the statements make sense (e.g., all objects, all people, etc.), and give a brief explanation:

5.4. If the original statement " $p \rightarrow q$ " is true, is the converse " $q \rightarrow p$ " always true? (Provide a counterexample if not.)

5.5. If " $p \rightarrow q$ " is true, is the contrapositive " $\neg q \rightarrow \neg p$ " always true?

5.6. If " $p \rightarrow q$ " is false, what can you say about the contrapositive " $\neg q \rightarrow \neg p$ "? What about the converse " $q \rightarrow p$ "?

Application

5.7. *Debugging logic*: A student writes, "If a figure is a square, then it has four sides. Therefore if a figure has four sides, it must be a square." Identify the student's error in terms of converse/contrapositive/inverse.

5.8. Think of a real-world conditional statement (not used in the text). State its converse and contrapositive, and comment on their truth in that real context.

Next: In Part 6: The Biconditional, we'll explore "if and only if" statements, which effectively combine an implication and its converse when both are true.

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7: The Biconditional

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Part 6: The Biconditional

Why Biconditionals Matter

"**It is daytime if and only if it is not nighttime.**" This statement does more than just "if it's daytime, then it's not nighttime" — it also says the reverse: "if it's not nighttime, then it's daytime." Whenever two statements always share the same truth value (either both true or both false), we connect them with a **biconditional** to express their equivalence.

Two Keys for One Lock

Imagine two different keys that happen to open the **same** lock. If you have Key A, you can open the lock; if you have Key B, you can **also** open the lock. If neither key is present, the lock stays closed. A biconditional captures this spirit of "**goes both ways**" — each statement guarantees the other.

What Is a Biconditional?

A **biconditional** joins two statements (p and q) and asserts "they are true or false together." We write it as:

$$p \leftrightarrow q,$$

and read it as:

┆ "*p if and only if q.*"

Often "if and only if" is abbreviated as "**iff**" in writing.

Everyday Examples

1. Daytime vs. Nighttime

2. p : "It is daytime."

3. q : "It is not nighttime."

$p \leftrightarrow q$: "It is daytime if and only if it is not nighttime."

Here both statements describe the same situation: either it's daytime (and thus not nighttime), or it's not daytime (meaning it is nighttime). They stand or fall together.

4. Half-empty vs. Half-full

5. p : "The cup is half empty."

6. q : "The cup is half full."

In many contexts, people treat these as equivalent descriptions of the same volume of liquid. So $p \leftrightarrow q$ (the cup is half empty **iff** it is half full) would be true (assuming consistent definitions).

7. **Six Eggs vs. a Half-Dozen Eggs**

8. p : "There are six eggs."

9. q : "There are a half-dozen eggs."

These mean exactly the same quantity. So $p \leftrightarrow q$ is true — whenever one is true, the other is true, and if one is false, the other is false.

10. **Access Control**

11. p : "You are authorized to enter the building."

12. q : "Your security badge is valid."

$p \leftrightarrow q$: "You are authorized to enter the building if and only if your security badge is valid."

This means having authorization and having a valid badge go hand-in-hand: Authorization requires a valid badge, and a valid badge guarantees authorization.

"If and only if" $\rightarrow$ "iff"

In many math and logic texts, "if and only if" is shortened to "**iff**." For example, one might write: "A shape is a square **iff** it has four equal sides and four right angles." (We won't rely on "iff" in this course, but it's useful to recognize.)

How a Biconditional Works

Two Directions in One Statement

Saying " p **if and only if** q " is equivalent to saying **both** of the following: 1. $p \rightarrow q$ (if p is true, then q is true)

2. $q \rightarrow p$ (if q is true, then p is true)

In symbols:

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

So $p \leftrightarrow q$ is true exactly when *both* implications hold (each implies the other).

Another way to phrase $p \leftrightarrow q$ is: " p is true **exactly when** q is true."

Truth Table for Biconditionals

Since a biconditional means " p and q share the same truth value," the truth table is:

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

We see $p \leftrightarrow q$ is true **only** in the first and last rows: when both p and q are T, or both are F. In the middle cases (T/F or F/T), the biconditional is false.

In summary, a biconditional is true when **both** p and q are true, or **both** are false.

Alternate View

Another way to say " p iff q " is that **either both are true or both are false**. In logical form, one can express this as:

$$\vee(p \wedge q) \vee (\neg p \wedge \neg q), \vee$$

which is logically equivalent to $p \leftrightarrow q$.

Why This Matters

1. **Mathematical Definitions:** Many mathematical statements use "if and only if" to state definitions or characterizations. For example, "A number is even **if and only if** it is divisible by 2." This guarantees that the condition is both necessary and sufficient. Recognizing biconditionals helps in understanding and constructing precise definitions.
2. **Full Logical Equivalence:** When you prove $p \leftrightarrow q$, you've shown that p and q are basically saying the same thing in different ways. This can simplify complex problems by allowing substitution of one for the other.
3. **Computer Science:** In programming, a biconditional is like an equality check between two boolean conditions (e.g., checking if two flags have the same value). It's used in algorithms and circuits where you need to ensure two conditions match.
4. **Problem Solving:** Knowing two conditions are equivalent (via a biconditional) means you can switch perspectives. For example, solving a puzzle might become easier if you replace a condition with an equivalent one that's easier to verify.

Necessary and Sufficient

- Saying " p is **necessary** for q " means $q \rightarrow p$ (if q happens, p must be true).
- Saying " p is **sufficient** for q " means $p \rightarrow q$ (if p happens, that's enough to ensure q).
- Saying " p is **necessary and sufficient** for q " is exactly the same as $p \leftrightarrow q$.

Common Misconceptions

1. "Any if-then can be made into an iff."

Reality: Only when both the statement and its converse are true does an *if and only if* hold. " $p \rightarrow q$ " by itself is not the same as " $p \leftrightarrow q$."

2. "Biconditional means a cause-effect works both ways."

Reality: Logic only cares about truth relationships, not causality. " p iff q " just means they are true together or false together. In real life, one might still cause the other or they might both depend on a common cause.

3. "Iff just means two statements are both true."

Reality: Not exactly — it means either both true or both false. It's about matching truth values, not just happening to be true at the same time.

Exercises

Rewriting as Biconditionals

Rewrite each statement explicitly in "if and only if" form (identify two directions):

- 6.1. "The person who gets the most votes is the winner." (Express as: winning the election $\leftrightarrow$ getting the most votes.)
- 6.2. "Thanksgiving is on the fourth Thursday of November." (Express as a biconditional defining when Thanksgiving occurs.)
- 6.3. "Going a mile is the same as going 5280 feet." (Express distance equivalence with iff.)

Identifying Directions

For each biconditional, break it into its two one-way implications ($\rightarrow$):

- 6.4. "Monique likes Jasper **if and only if** Jasper likes Monique."
- 6.5. "The light is on **if and only if** the switch is up."

(For each above, write the $p \rightarrow q$ form and the $q \rightarrow p$ form.)

True or False Biconditionals

Assume the following facts for a particular universe:

- "Trees have wings" = False
- "Mountains have ears" = False
- "Mountains have trees" = True
- "Wings have feathers" = True

Determine which of these biconditionals are true and which are false in that universe, and explain why based on matching truth values:

- 6.6. "Trees have wings **if and only if** wings have feathers."
6.7. "Mountains have ears **if and only if** trees have wings."
6.8. "Mountains have trees **if and only if** wings have feathers."

Changing Truth Values

For " $p \leftrightarrow q$," explain what happens in each scenario (to the truth of the biconditional):

- 6.9. p is made the opposite truth value (flipped from T to F or F to T) while q stays the same.
6.10. Both p and q are made the opposite truth values simultaneously.

(Does the biconditional remain true, become false, or become undetermined? Use examples to illustrate.)

Logical Puzzles with Biconditionals

- 6.11. Create a **real-life biconditional** you believe is true. Then break it into the two directional statements. Are both directions obviously true in normal reasoning? (For example, "You can access the website if and only if you have the correct password" — check both directions.)
- 6.12. Find (or invent) a statement that **appears** to be a biconditional but isn't always true. Explain where the equivalence breaks down. (For example, "A person is rich if and only if they are happy" — identify if either direction fails.)
- 6.13. Provide a scenario in science or math where stating an "if and only if" condition changes the meaning of a claim. Discuss why requiring both directions (necessity and sufficiency) is important in that context.
- 6.14. **Puzzle:** Three friends each make one statement, and exactly one of the three statements is true: - Alice: "Bob is lying if and only if I am telling the truth."
- Bob: "Carlos is telling the truth if and only if Alice is lying."
- Carlos: "Alice is telling the truth if and only if Bob is telling the truth."

Who is telling the truth and who is lying? (Hint: Translate each statement into logical conditions about who is truthful (T) or lying (F), then find a consistent assignment where exactly one is T.)

Next: In Part 7: Representing Statements Symbolically, we'll look at how to translate complex statements into logical notation and ensure they're interpreted correctly (introducing formal notation and precedence rules).

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8: Representing Statements Symbolically

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Part 7: Representing Statements Symbolically

Why Symbolic Representation Matters

Think of complex logic like a puzzle: you have pieces (negation, conjunction, disjunction, conditional, biconditional) and you need to arrange them to see the full picture of an argument. **Symbolic representation** is like having a **clear blueprint** for how those puzzle pieces fit together. When statements get complicated, using well-defined symbols and following rules of precedence helps us **avoid confusion** and **spot logical relationships** easily.

Architect's Blueprint

Building an argument is like constructing a house:

- You have bricks (basic propositions) and different ways to join them (connectives).
- A detailed **blueprint** ensures you build in the correct order and structure.
- Without a clear diagram, you risk confusion about which part connects where. Symbolic logic provides that clear diagram for reasoning.

Our Five Propositional Connectives

So far, we have five logical connectives. Let's summarize them with their symbols:

Symbol	Meaning	Name
$\neg$	not	negation
$\wedge$	and	conjunction
$\vee$	or	disjunction
$\rightarrow$	implies	conditional
$\leftrightarrow$	if and only if	biconditional

Note: In different texts you might see "&" instead of $\wedge$, or "v" instead of $\vee$. In this course we'll stick to the symbols above for consistency.

Rules of Precedence and Parentheses

When writing logical expressions, parentheses clarify **which operations apply first**. Similar to arithmetic (where multiplication happens before addition unless parentheses dictate otherwise), certain logical connectives bind more tightly.

To avoid ambiguity, we use these **precedence conventions** (from highest precedence to lowest):

1. Negation ($\neg$) – evaluate this first.

Example: $\neg p \wedge q$ means $(\neg p) \wedge q$, **not** $\neg(p \wedge q)$. If you meant the latter, you'd need parentheses.

2. Conjunction ($\wedge$) and Disjunction ($\vee$) – evaluate these next, before implications. They share the same precedence level.

Example: $p \vee q \rightarrow r$ is interpreted as $(p \vee q) \rightarrow r$, **not** $p \vee (q \rightarrow r)$. If your intent was the latter, parentheses are required.

3. Conditional ($\rightarrow$) and Biconditional ($\leftrightarrow$) – these have the lowest precedence and are evaluated last (unless parentheses dictate otherwise). They are not associative, meaning you can't just drop parentheses if you chain them.

Additionally, some connectives are **associative**, meaning the grouping doesn't matter for consecutive occurrences: - **Associative:** $\wedge$ and $\vee$ are associative. $(p \wedge q) \wedge r$ is equivalent to $p \wedge (q \wedge r)$, so we can write $p \wedge q \wedge r$ unambiguously. Likewise for $p \vee q \vee r$.

- **Non-Associative:** $\rightarrow$ and $\leftrightarrow$ are **not** associative. $(p \rightarrow q) \rightarrow r$ is generally not the same as $p \rightarrow (q \rightarrow r)$. We **always** need parentheses when chaining multiple $\rightarrow$ or $\leftrightarrow$ in one expression to avoid confusion.

Analogy with Arithmetic

- Multiplication binds tighter than addition (just like $\wedge$ or $\vee$ bind tighter than $\rightarrow$).
- A unary minus (like $-x$) has high precedence (like $\neg$ does in logic).
- You can drop parentheses for $(x + y) + z$, but not for $(x + y) \cdot z$ without changing meaning. Similarly, we can drop some parentheses with $\wedge$ or $\vee$, but not around $\rightarrow$ or $\leftrightarrow$.

Examples: When to Omit or Use Parentheses

Let's practice applying these rules:

1. Negation Precedence

$\neg p \wedge q$ means $(\neg p) \wedge q$.

If you wanted $\neg(p \wedge q)$, you **must** write the parentheses.

2. $\wedge$ and $\vee$ over $\rightarrow$

$p \rightarrow q \wedge \neg r$ by precedence means $p \rightarrow (q \wedge (\neg r))$.

If you wrote $p \rightarrow q \wedge \neg r$ without thinking, it's read as $p \rightarrow (q \wedge \neg r)$. To express $(p \rightarrow q) \wedge \neg r$, add parentheses: $(p \rightarrow q) \wedge \neg r$.

3. Dropping parentheses for consecutive $\wedge$ or $\vee$

$(p \vee q) \vee r$ can be simply written $p \vee q \vee r$ (since $\vee$ is associative).

But $(p \rightarrow q) \rightarrow r$ cannot drop parentheses because it would mean something different if grouped differently.

4. Non-Associativity of $\rightarrow$

$(p \rightarrow q) \rightarrow r$ vs $p \rightarrow (q \rightarrow r)$: these can have different truth values (Exercise 7.17 will have you examine this difference).

Always use parentheses to clarify in such cases.

Common Mistake

Dropping parentheses around implications ($\rightarrow$) or biconditionals ($\leftrightarrow$) is a frequent error. It can completely change the meaning of an expression. When in doubt, put parentheses to enforce the intended grouping.

Real-Life Angle: Why Bother With All These Rules?

- **Streamlined Communication:** Just as you wouldn't write a math expression with every possible parenthesis (that would be cluttered), you don't want to clutter logical expressions either. Precedence rules let us omit *unnecessary* parentheses, making statements easier to read while still being clear.
- **Reduced Ambiguity:** By everyone following the same conventions, **anyone** reading the expression interprets it the same way. This avoids miscommunication in proofs or program code.
- **Avoiding Logical Errors:** In programming, the precedence of operators can change how code runs. Similarly in logic, forgetting a needed parenthesis around an implication can lead you to interpret a statement incorrectly and draw wrong conclusions.

Common Misconceptions

- "All parentheses are optional in logic; it's all about context."

Reality: Only *some* parentheses can be safely dropped under specific precedence or associativity rules. Others are essential to preserve meaning.

- " $\wedge$ and $\vee$ work like $\rightarrow$, so we can't drop any parentheses with them."

Reality: $\wedge$ and $\vee$ are associative, so you can drop parentheses in a chain of all ANDs or all ORs. Conditionals ($\rightarrow$) and biconditionals ($\leftrightarrow$) are not associative, so you cannot drop those parentheses.

- " $(p \vee q) \rightarrow r$ means the same as $p \vee (q \rightarrow r)$."

Reality: They are generally not the same. For instance, if p is true, $(p \vee q) \rightarrow r$ only cares about r for truth value, whereas $p \vee (q \rightarrow r)$ will be true regardless of r because p is true (no conditional in that form). They differ.

Exercises

Symbolic Representation

For each of the following English statements, translate into **symbolic logic** using the proper symbols and keeping the intended meaning clear. Introduce letters (e.g., p, q, r, s) for the component propositions, and use logical connectives ($\neg, \wedge, \vee, \rightarrow, \leftrightarrow$) as appropriate:

- 7.1. "If it rains in the forest, the trees get wet."
- 7.2. "If you pass the test, then you won't fail the course, and if you fail the course, you will have to go home."
- 7.3. "Bob will sit next to Alice if and only if Jose does not sit next to Darren."
- 7.4. "If a tree falls in the forest and no one is there to hear it, then it makes no sound."
- 7.5. "If the slithy toves did gyre and gimble in the wabe, then either the mome raths were not outgrabe or it was brillig."
- 7.6. "If I don't finish my homework tonight, then either I will have to finish it tomorrow or I will get a bad grade; but if I **do** finish my homework tonight, then I will miss my favorite TV show."
- 7.7. "I will win the chess game if I checkmate my opponent, if she resigns, or if she runs out of time." (Hint: restructure as a logical OR of conditions implying your win.)
- 7.8. "I will declare a major in astronomy, and I will declare a minor in either philosophy or anthropology."
- 7.9. "If it is sunny outside, then I will go to the beach if I have some free time."

Removing Parentheses

Apply the **precedence** and **associativity** rules to eliminate parentheses where it's valid. **Do not** remove parentheses if they're necessary for the original meaning:

- 7.10. $(a \wedge (\neg b)) \leftrightarrow (c \vee d)$ (Hint: Can any outer parentheses be dropped or re-grouped?)
- 7.11. $(\neg(a \vee b)) \leftrightarrow ((\neg a) \wedge (\neg b))$ (This is a classic equivalence — see if you can simplify the notation.)
- 7.12. $((a \rightarrow b) \wedge (b \rightarrow a)) \leftrightarrow (a \leftrightarrow b)$
- 7.13. $((a \rightarrow b) \vee (b \rightarrow a)) \vee (a \wedge (\neg a))$
- 7.14. $\neg(a \leftrightarrow c) \wedge \neg(b \leftrightarrow c)$
- 7.15. $(a \vee b) \wedge (c \wedge d) \wedge (b \vee c) \wedge a$ (Use associativity and commutativity of $\wedge$ and $\vee$ as needed.)

Showing Logical Distinctions

- 7.16. Show with specific truth values for p, q , and r that $(p \vee q) \wedge r$ and $p \vee (q \wedge r)$ are **not logically equivalent** (i.e. find a counterexample where one is true and the other is false).
- 7.17. Similarly, find truth values for p, q , and r such that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ have different truth values, demonstrating they are not equivalent expressions.

Extra Practice

- 7.18. Invent a complex statement (using at least four distinct propositions and multiple connectives). Write it out with **full parentheses** showing every grouping. Then rewrite it in a cleaner way by removing any unnecessary parentheses using the rules learned. Ensure the meaning stayed the same.

Next: In **Part 8: More on Truth Tables**, we'll deepen our understanding by analyzing complex statements via truth tables and exploring concepts like tautologies, contradictions, and logical equivalences.

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9: More on Truth Tables

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Part 8: More on Truth Tables

Why Understanding Truth Tables Matters

Throughout our journey, we've used **truth tables** as a reliable way to see when a compound statement is **true** or **false**. Think of them as **blueprints** showing how different combinations of truth values for propositions affect an overall expression.

A Map of Every Possibility

Imagine driving through a city where **every possible turn** (true/false decision) must be considered. A truth table is like having a complete map: you check every route (combination of truths and falses) so you never miss a scenario where your expression might change from true to false or vice versa.

Recap: The Five Propositional Connectives

We've encountered five key logical connectives. Here are their **standard truth tables** for reference:

Negation ($\neg$)

p	$\neg p$
T	F
F	T

Conjunction ($p \wedge q$)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \wedge q$
F	F	F

Disjunction ($p \vee q$)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional ($p \rightarrow q$)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional ($p \leftrightarrow q$)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

These tables are the building blocks for constructing truth tables of more complex expressions.

Building and Reading Truth Tables

Complete Truth Table

A **complete** truth table explicitly lists **all** intermediate columns for sub-expressions, building up to the final result. You construct it step by step:

1. **List all possible truth combinations** of your basic propositions ($p, q, r, \dots$). For n basic propositions, you'll have 2^n rows covering every combination of T/F.
2. **Compute intermediate columns** for smaller parts of the expression ($\neg p, p \vee q$, etc.) as needed.
3. **Combine** these intermediate results to determine the truth of the final compound expression in each row.

Example: $p \wedge q \leftrightarrow \neg(\neg p \vee \neg q)$

Let's do a complete table for the expression:

$$(p \wedge q \leftrightarrow \neg(\neg p \vee \neg q))$$

We suspect this might always be true (it looks like a known logical equivalence: the right side resembles the logical NOT of an OR, which by De Morgan's law equals $p \wedge q$). The truth table will confirm if it's always true (a tautology).

We'll create columns for each sub-expression: $p, q, p \wedge q, \neg p, \neg q, \neg p \vee \neg q, \neg(\neg p \vee \neg q)$, and finally $p \wedge q \leftrightarrow \neg(\neg p \vee \neg q)$.

p	q	$p \wedge q$	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$\neg(\neg p \vee \neg q)$	$p \wedge q \leftrightarrow \neg(\neg p \vee \neg q)$
T	T	T	F	F	F	T	T

p	q	$p \wedge q$	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$\neg(\neg p \vee \neg q)$	$p \wedge q \leftrightarrow \neg(\neg p \vee \neg q)$
T	F	F	F	T	T	F	T
F	T	F	T	F	T	F	T
F	F	F	T	T	T	F	T

Notice the final column is **T in every row**. This shows:

$$(p \wedge q \equiv \neg(\neg p \vee \neg q))$$

which is indeed a tautology (they are equivalent for all possible truth values of p and q). This is a formal statement of one of **De Morgan's laws** (the negation of an OR is the AND of the negations).

Condensed Truth Table

Sometimes we don't need to show every intermediate step. A **condensed** table omits columns for sub-expressions that we don't need to explicitly see. We still include all combinations of basic propositions, but we calculate only the parts we're interested in.

Condensed Table Example

Suppose we want to analyze the expression $p \wedge \neg q \rightarrow \neg r$. Instead of writing out every possible sub-expression, we might keep just enough columns to reach the final result. With three propositions p, q, r , there are $2^3 = 8$ rows. We might choose columns for p, q, r , then $\neg q$, then $p \wedge \neg q$, then $\neg r$, and finally $p \wedge \neg q \rightarrow \neg r$:

p	q	r	$\neg q$	$p \wedge \neg q$	$\neg r$	$p \wedge \neg q \rightarrow \neg r$
T	T	T	F	F	F	T
T	T	F	F	F	T	T
T	F	T	T	T	F	F
T	F	F	T	T	T	T
F	T	T	F	F	F	T
F	T	F	F	F	T	T
F	F	T	T	F	F	T
F	F	F	T	F	T	T

We didn't include separate columns for (for example) $p \wedge \neg q \rightarrow \neg r$'s subparts separately (like we combined computing the implication in one go at the end). The final column can be filled in by reasoning or by computing mentally using earlier columns. This is more efficient than a fully expanded table, but relies on your understanding to fill in some steps.

Deeper Look: Tautologies, Contradictions, and Contingencies

An expression that's **true in every row** of its truth table is called a **tautology** (or is *logically valid*). If it's false in every row, it's a **contradiction**. If it's true in some rows and false in others, it's a **contingency**.

In our complete table example above, the final column was all T, indicating a tautology. Let's summarize:

- **Tautology:** True under **all** possible truth assignments to its basic propositions. (Example: $p \vee \neg p$ is always true, often called the Law of Excluded Middle.)
- **Contradiction:** False under **all** possible assignments (the final column would be all F). (Example: $p \wedge \neg p$ is always false.)
- **Contingency:** True in **some** cases and false in **others**.

Example of a Tautology

"It's raining or it's not raining." In logical form: $p \vee \neg p$. No matter the truth value of " p " ("It's raining"), the statement is **always** true. This is a tautology.

Examples in Each Category

- **Tautology Example:** $p \vee \neg p$ (the Law of Excluded Middle). For any truth value of p , either p is true or $\neg p$ is true. The table would be T in all rows.
- **Contradiction Example:** $p \wedge \neg p$. This is always false — p cannot be both true and false at the same time.
- **Contingency Example:** $p \rightarrow q$. As we've seen, $p \rightarrow q$ can be true or false depending on p and q . Some rows of its truth table are true (when p is false or q is true), and one row is false (when p is true and q false).

Applications of Truth Tables (Continued)

Truth tables are not just academic; they play a role in various domains:

Validating Arguments (using truth tables)

As mentioned, you can use truth tables to test argument validity by checking all cases. If you append the premises and conclusion in a table and find *no row* where premises are all true and conclusion false, the argument is valid (truth-table method for validity).

Circuit Design (revisited)

When designing a logical circuit (say for an electronics project or a computer chip), you might start with a desired truth table (behavior) and then derive a circuit that matches it. Conversely, for a given circuit, you can derive the truth table to understand its behavior fully.

Software Testing (exhaustive testing)

Consider conditions in an `if` statement with multiple clauses combined by AND/OR. A truth table of those conditions helps ensure you've tested every critical combination. For instance, if a decision depends on three booleans, there are 8 combinations to test; a truth table enumerates them clearly.

Legal and Policy Analysis

Complex regulations often involve multiple conditions (if X and Y but not Z, then outcome W, etc.). Making a truth table of conditions (where T/F indicates whether a condition is met) can help spot if there are any scenarios not accounted for or if some rule is redundant or contradictory.

Example: Complex Condition Analysis

Policy Logic Check

A policy states: "An employee gets a bonus if they have worked at the company for more than 5 years or (they have worked at least 2 years and made a significant achievement), and they are not under any disciplinary action."

Let:

- p = "Worked > 5 years"
- q = "Worked ≥ 2 years AND made significant achievement"
- r = "Under disciplinary action"

The rule can be expressed as: $(p \vee q) \wedge \neg r$.

To ensure this policy makes sense, a truth table for $(p \vee q) \wedge \neg r$ could be examined. It will show, for instance, that if an employee has *not* worked ≥ 2 years (so p and q are false), then even if they're not under discipline, they won't get a bonus (because the experience/achievement condition fails). The table can reveal if any combination seems unfair or unintended, prompting a policy revision.

Common Misconceptions (Advanced)

- **"If the final column isn't all T or all F, the statement must be random."**
Reality: A mix of T and F in the final column (contingency) often still has *patterns*. For instance, $p \rightarrow q$ is false in only one specific scenario. Recognizing those patterns is key to understanding logical relationships (like equivalences and implications between statements).
- **"Two different-looking expressions can't have the same truth table."**
Reality: They can! As we saw, $p \wedge q$ and $\neg(\neg p \vee \neg q)$ had identical truth tables (hence they are equivalent). Part of advanced logical analysis is identifying when different expressions actually mean the same thing logically.

- "To check two expressions are equivalent, I have to compare them in general, I can't just use one truth table."
Reality: You can create a single truth table with both expressions side by side. If their columns match for every row, they are logically equivalent. This truth-table method is a sure-fire way (though it can be tedious for many variables).

Exercises

- 8.1. Construct a truth table for $(p \wedge r) \vee (\neg p \wedge \neg r)$. Is this a tautology, contradiction, or contingency? (Hint: Compare to the biconditional truth table form.)
- 8.2. Determine via truth table whether $(p \rightarrow r) \wedge (q \rightarrow r)$ is equivalent to $(p \vee q) \rightarrow r$. (This asks if having two separate conditions implying r is the same as their disjunction implying r .)
- 8.3. Use a truth table to check if $\neg p \vee (p \wedge q)$ is equivalent to $p \wedge q$. (This can tell you if a certain logical simplification is valid.)
- 8.4. Provide a brief real-world interpretation of a tautology and a contradiction (in terms of a statement or rule that is always true or always false).
- 8.5. Explain why a tautology can never be false (by referencing the truth table concept). Also, explain why a contradiction can never be true.
- 8.6. Imagine a simple electrical circuit that has two switches (A and B) in parallel controlling a light (meaning the light is on if either switch is on). Write the logical expression for the light being on in terms of A and B (use, say, A for "Switch A is ON"). Create a truth table for this expression and identify it as tautology, contradiction, or contingency.
- 8.7. (Challenge) We have three conditions in a board game that determine if a player gets a bonus: p = has the most points, q = found the secret item, r = helped another player. The rule is: "The player gets a bonus if either (p is true, or both q and r are true), but not if all three conditions are true." Express this rule in a logical formula and then create a truth table for it. Identify any contingencies (rows where sometimes they get a bonus, sometimes not, depending on conditions).
- 8.8. Provide a real-life scenario or statement that corresponds to a tautology. For instance, "Either I will go to the party or I will not go to the party." Then do the same for a contradiction. Explain why each fits the definition.

With these truth table skills, you can confidently analyze even complex logical statements and arguments. The next parts of the course will introduce **Predicate Logic** (where we talk about "all" and "some") and other advanced topics, building on the foundation we've established in propositional logic.

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10: Predicate Logic

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Part 9: Predicate Logic

Why Predicate Logic Matters

Propositional logic treats entire statements as indivisible units (propositions). But often we want to talk about **properties of objects** or **relationships between objects**. For example, "All cats are mammals" or "Some numbers are prime." Propositional logic can't easily express "all" or "some" — it would just see "All cats are mammals" as a single black box statement. **Predicate logic** (also known as *first-order logic*) allows us to peek inside that box. It lets us use variables and quantifiers to say things about **many instances at once**, greatly expanding the expressiveness of our logic.

In simpler terms, predicate logic answers questions like: - How do we say *every* member of a category has some property?
- How do we say *there exists* something with a certain property?

These abilities are crucial in mathematics, computer science (database queries, AI), and philosophy.

From Aristotle to Modern Logic

Aristotle's syllogisms (e.g., "All men are mortal; Socrates is a man; therefore Socrates is mortal") were an early form of predicate logic dealing with categories and membership. Modern predicate logic (developed by Frege in 1879 and others) formalized these ideas with quantifiers and variables, laying the groundwork for much of modern mathematics and computer science.

From Propositions to Predicates

A **predicate** is like a property or relation that can be true or false of something. It's often represented as a capital letter with parentheses, e.g. $P(x)$ might mean "x is prime" or "x is a philosopher", depending on context. The variable x can stand for an object in the domain of discourse (the set of things we're talking about).

Example: Let $M(x)$ mean "x is mortal" and $H(x)$ mean "x is human." The statement "All humans are mortal" can be expressed in predicate logic, whereas propositional logic could not directly capture it.

Quantifiers: Universal and Existential

Predicate logic introduces **quantifiers** to indicate how many or which elements of the domain we're talking about:

- **Universal Quantifier** ($\forall$): Means "for all" or "for every".
 $\forall x P(x)$ says "for all x , $P(x)$ is true." (Every x has property P .)
- **Existential Quantifier** ($\exists$): Means "there exists" or "for at least one".
 $\exists x P(x)$ says "there exists at least one x such that $P(x)$ is true." (Some x has property P .)

Let's see them in use:

"All" statements (Universal)

- **All cats are mammals.**

Domain: animals.

Let $C(x)$ = "x is a cat", $M(x)$ = "x is a mammal."

In logic: $\forall x (C(x) \rightarrow M(x))$.

This says: for every creature x , if x is a cat, then x is a mammal.

- **No dogs can fly.**

Let $D(x)$ = "x is a dog", $F(x)$ = "x can fly."

"No dogs can fly" can be written as: $\forall x (D(x) \rightarrow \neg F(x))$.

(For every x , if x is a dog, then x cannot fly.)

Alternatively, one could write $\neg \exists x (D(x) \wedge F(x))$ meaning "there does not exist an x that is a dog and can fly," which is logically equivalent.

"Some" statements (Existential)

- **There exists a prime number greater than 100.**

Domain: numbers.

Let $P(x)$ = "x is prime", $G(x)$ = "x > 100\$".

In logic: $\exists x (P(x) \wedge G(x))$.

(There is at least one number x such that x is prime and $x > 100$.)

- **Someone in the class speaks French.**

Domain: people (in the class).

Let $F(x)$ = "x speaks French."

$\exists x F(x)$ says "There is at least one person x such that x speaks French."

Combining Quantifiers and Logic

We can make more complex statements:

- **"Every even number greater than 2 is not prime."**

Domain: natural numbers.

$E(x)$ = "x is even", $P(x)$ = "x is prime", $G(x)$ = "x > 2\$".

Logic: $\forall x ((E(x) \wedge G(x)) \rightarrow \neg P(x))$.

(For any number, if it's even and >2, then it is not prime.)

- **"There is a student who is in both the chess club and the debate club."**

Domain: students.

$C(x)$ = "x is in the chess club", $D(x)$ = "x is in the debate club".

$\exists x (C(x) \wedge D(x))$.

(At least one student is in both clubs.)

Translating 'none' or 'no one'

Statements like "No x satisfy condition Y" can be translated in two equivalent ways:

- **Universal form:** $\forall x$ (if $Y(x)$ then false) – e.g., $\forall x (Y(x) \rightarrow \text{false})$, which simplifies to $\forall x \neg Y(x)$ (if we mean no x at all have property Y). But more typically, "No X are Y" is translated as $\forall x (X(x) \rightarrow \neg Y(x))$.

- **Negated existential form:** $\neg \exists x Y(x)$ – "There does not exist an x with property Y."

These are equivalent: saying no x has property Y is the same as saying "for all x , x does not have property Y."

Domains and Scope of Quantifiers

When we write $\forall x$ or $\exists x$, we presume a **domain of discourse** for x (the set of all things x could refer to). Sometimes it's stated explicitly ("for all people x , ..."), other times it's implied by context (if we're talking about numbers, x ranges over numbers).

Quantifiers have a **scope** similar to how parentheses or logical operators have scope. For example, in $\forall x (P(x) \vee Q(y))$, the $\forall x$ applies only to the $P(x)$ part (and not to $Q(y)$ which might involve a different variable).

If we need multiple quantifiers, we can use different variables or nest them:

- **"Every student admires some professor."**

Domain: people at a university.

Let $S(x)$ = "x is a student", $Prof(y)$ = "y is a professor", $A(x, y)$ = "x admires y".

Logic: $\forall x [S(x) \rightarrow \exists y (Prof(y) \wedge A(x, y))]$.

(For each person x, if x is a student, then there exists at least one person y who is a professor and x admires y.)

- **Order matters** for different quantifiers:

$\forall x \exists y A(x, y)$ is not the same as $\exists y \forall x A(x, y)$. The first says "for each x , you can find a (potentially different) y such that $A(x, y)$." The second says "there is a single y that works for every x ." This distinction is huge.

Order of Quantifiers

Pay attention to quantifier order:

- $\forall x \exists y P(x, y)$: For each x , *maybe different* y s can witness the truth.

- $\exists y \forall x P(x, y)$: There is *one particular* y that works for all x .

These can mean very different things. For example, "For every person, there is someone whom they love" versus "There is a person whom everyone loves."

Translating Between English and Predicate Logic

It takes practice to capture English precisely in symbols:

- *Each, every, all* $\rightarrow \forall$
- *Some, there is/are, at least one* $\rightarrow \exists$
- *None, no* $\rightarrow$ use $\forall$ with a negation, or $\neg \exists$

Example translations:

- "Every car has wheels."

Domain: vehicles (or things).

$C(x)$ = "x is a car", $W(x)$ = "x has wheels".

$\forall x (C(x) \rightarrow W(x))$.

- "Some cars have no wheels."

$C(x)$ = "x is a car", $W(x)$ = "x has wheels".

$\exists x (C(x) \wedge \neg W(x))$.

(There exists an object that is a car and does not have wheels — perhaps this is false in reality, but that's the logical form.)

- "Not every car has wheels."

This is the negation of "every car has wheels."

We can write: $\neg \forall x (C(x) \rightarrow W(x))$, which is equivalent to $\exists x \neg (C(x) \rightarrow W(x))$ which simplifies to $\exists x (C(x) \wedge \neg W(x))$.

So "Not all have wheels" means "Some car has no wheels" — note how English "not every" turns into an existential statement logically.

- "At least two people have the same birthday." (A bit trickier)

Domain: people.

$B(x, d)$ = "Person x has birthday on date d ".

One way: $\exists x \exists y \exists d [(x \neq y) \wedge B(x, d) \wedge B(y, d)]$.

(There exist two different people x and y and a date d such that x and y both have birthday d .)

Real-World Applications of Predicate Logic

- **Database Queries:** SQL and other query languages are built on predicate logic. A query like "SELECT * FROM Employees WHERE age > 30 AND department = 'Sales'" is basically finding all x such that $Age(x) > 30$ and $Dept(x, 'Sales')$.
- **Artificial Intelligence:** Knowledge representation often uses first-order logic to encode facts and rules. E.g., in an expert system: $\forall x (Cat(x) \rightarrow Mammal(x))$ encodes a general fact about the world.

- **Mathematics:** Almost all definitions and theorems in mathematics are expressed in predicate logic ("For every $\varepsilon > 0$, there exists a $\delta > 0$ such that ...", which is a $\forall\exists$ statement).
- **Formal Verification:** When verifying hardware or software, one often writes specifications like "for all possible inputs satisfying condition P, the output will satisfy Q" ($\forall x(P(x) \rightarrow Q(x))$) and uses automated theorem provers to check it.

Common Misconceptions

- " $\forall x P(x)$ means the same as $P(x)$ for a generic x ."
Reality: $\forall x P(x)$ means *every possible* x in the domain makes $P(x)$ true. You can't assume a specific one without stating it. It's a statement about the entire domain at once.
- " $\exists x P(x)$ asserts the existence but we know nothing else about that x ."
Reality: Yes, existential just guarantees at least one exists, but we usually can't give it a name unless we use a witness. It's a logical guarantee without identification (unless the context or proof finds a concrete example).
- "If $\forall x P(x)$ is true, then $\exists x P(x)$ must be true."
Reality: True. If everything has property P, certainly at least one thing does. (In logic, $\forall x P(x)$ implies $\exists x P(x)$ *provided the domain is nonempty*. Usually we assume domains are nonempty.)
- "If $\exists x P(x)$ is true, then $\forall x P(x)$ might be true."
Reality: Having one example doesn't mean all have it. E.g., "There exists a number that is even" is true, but "For all numbers, they are even" is false.
- **Mixing up $\forall$ and $\exists$:** It's common but critical to keep them straight. "Every" vs "Some" are not interchangeable.

Exercises

- 9.1. Translate into predicate logic: "All birds can fly." (Define appropriate predicates for being a bird and being able to fly.)
- 9.2. Translate: "Some birds cannot fly." (You might know real examples, but just do it logically.)
- 9.3. Translate: "Every student in this class has taken at least one programming course." (Define predicates like $S(x)$ for "x is a student in this class", and $P(x, y)$ for "x has taken course y", and maybe a set or predicate for programming courses.)
- 9.4. Translate: "There is a prime number that is even." (Remember what the only even prime is!)
- 9.5. Consider the statement: "If a number is rational, then it can be written as a ratio of two integers." Express this as a logical statement with a universal quantifier. (Let $R(x)$ = "x is rational", and define a predicate for "x can be written as m/n with m,n integers".)
- 9.6. Express the negation of the statement in 9.5 in a normalized way (so that negation symbol directly precedes the predicate, not the quantifier). Explain the meaning of the negation in words.
- 9.7. Translate: "Not every book is interesting." Then contrast it with "No book is interesting." (They are different! Provide both translations.)
- 9.8. Translate: "Everyone has exactly one mother." (Hint: This one is a bit advanced. It means: for every person x , there exists a person y such that y is the mother of x , and for all z , if z is a mother of x then $z = y$.)
- 9.9. Is the following statement true or false (in ordinary arithmetic)? "For every integer y , there exists an integer x such that $x < y$." Translate it to logic as well, then answer.
- 9.10. Explain in your own words the difference between $\forall x \exists y P(x, y)$ and $\exists y \forall x P(x, y)$. Give a real-life example where one is true but the other is false.

Advanced: Predicate logic is a powerful tool, but with great power comes additional complexity. In further studies, you'll encounter concepts like relations, functions, multiple quantifiers, and even logical paradoxes that arise with unrestricted domains (like Russell's paradox about "the set of all sets that don't contain themselves"). Predicate logic is the gateway to these deeper explorations, including the foundations of mathematics and computer science.

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11: Modal Logic

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Part 10: Modal Logic

Why Modal Logic Matters

Up to now, our logical statements have been either true or false in an absolute sense. But often, we want to reason about **possibilities** and **necessities** rather than concrete truth. For example: - "It is **possible** that it will rain tomorrow."

- "Given the rules of the game, it's **necessary** that one team wins."

Modal logic extends classical logic by introducing modalities – concepts like **necessity** and **possibility** (and others like obligation, knowledge, time, etc., in various branches). Modal logic allows us to talk not just about what *is* true, but about what *could be* true or *must be* true under certain conditions.

Parallel Worlds

One way to think of modal logic is to imagine multiple "possible worlds."

- **Necessarily** p means p is true in **all** imaginable worlds (there's no scenario where p is false).

- **Possibly** p means p is true in **at least one** imaginable world (there is some scenario where p holds).

It's like having a map of different outcomes or universes, and modal statements tell us how a proposition fares across those universes.

Modal Operators: Necessity ($\Box$) and Possibility ($\Diamond$)

Modal logic introduces two new operators (we'll use the common symbols from modal logic):

- $\Box$ (**Necessarily**): $\Box p$ means " p is necessarily true." In other words, in every relevant scenario or world, p holds. We sometimes read $\Box p$ as "it must be that p ."
- $\Diamond$ (**Possibly**): $\Diamond p$ means " p is possibly true." In other words, in at least one possible scenario, p holds. Think of $\Diamond p$ as "it's not ruled out that p ," or "there is a chance/option that p ."

These operators are inter-definable by negation:

- $\Box p$ is logically equivalent to $\neg \Diamond \neg p$ (if p is necessary, then it's impossible for p to be false).

- $\Diamond p$ is equivalent to $\neg \Box \neg p$ (if p is possible, then it's not necessary that p is false).

Everyday Interpretations

- $\Box p$: " p is true in all cases" or " p is true no matter what."
Example: " $\Box(2 + 2 = 4)$ " is true (mathematically, $2+2=4$ is necessarily true; there's no scenario where it's false).
" $\Box(\text{the sky is blue})$ " – this is false, because we can imagine scenarios (times of day, other planets, etc.) where the sky is not blue.
- $\Diamond p$: " p might be true" or "it's possible that p ."
Example: " $\Diamond(\text{it will rain tomorrow})$ " – true (assuming it's not utterly impossible by some magic; there's a chance of rain).
" $\Diamond(2+2=5)$ " – false (there's no possible world in classical arithmetic where $2+2$ equals 5).

Combining Modal Operators with Logic

Modal operators work like an added layer on top of propositional logic. We can nest them or combine with connectives:

- $\Box p \wedge \Diamond q$: "It is necessarily the case that p is true, and it is possibly the case that q is true."
- $\Box(p \rightarrow q)$: "It is necessarily true that if p then q ." (In every possible world, if p then q holds.)
- $\Diamond(p \wedge q)$: "Possibly, both p and q are true at the same time."

We interpret these by imagining different scenarios (possible worlds) and the modal must hold across those worlds as specified.

Reasoning Patterns in Modal Logic

There are some common intuitions: - If $\Box p$ is true, then p is true (because if p is true in all worlds, in particular it's true in the actual world). Formally, $\Box p \rightarrow p$. This property is often assumed in modal logic systems (known as the T axiom). - If $p \rightarrow q$ is necessary, and p is necessary, then q is necessary. (If in every world $p \rightarrow q$, and in every world p , then in every world q .) Formally, $(\Box(p \rightarrow q) \wedge \Box p) \rightarrow \Box q$. - $\Diamond p$ does *not* imply p (just because something is possible doesn't mean it's actually true). And p does not imply $\Box p$ (just because it's true here doesn't mean it's necessarily always true).

Examples of Modal Reasoning

- **Physical Necessity**: "It is necessary, given the laws of physics, that nothing can travel faster than light." We might formalize: $\Box(\text{no object travels faster than light})$ relative to worlds with the same physical laws.
- **Logical Necessity vs Physical**: "Necessarily $2+2=4$ " (logical/mathematical necessity, true in all conceivable worlds) versus "Necessarily, the sun rises in the east" (this might be physically necessary if laws of nature are fixed, but one could imagine a different planet or axis flip — so maybe not absolutely necessary).
- **Possibility**: "It's possible I will win the lottery." $\Diamond(\text{I win lottery})$ is true (assuming I bought a ticket). "It's possible cats can talk" $\Diamond(\text{cats talk})$ might be false under our understanding of biology (or we consider a very different possible world? If we allow a far-fetched scenario, maybe it's logically possible, just not physically in our world).
- **Epistemic Modal (knowledge)**: Sometimes $\Box$ and $\Diamond$ are interpreted as "I know for sure" vs "for all I know, maybe". For example, "For all I know, it's possible that the lights are still on at home" would be $\Diamond(\text{lights on})$ in an epistemic sense. However, to avoid confusion, we'll stick with the basic necessity/possibility interpretation.

Real-World Applications of Modal Logic

- **Philosophy**: Modal logic was originally developed for philosophical reasoning (like necessity in metaphysics, possibility in counterfactuals, etc.). Also used in analyzing philosophical arguments (like modal versions of proofs or the famous "modal ontological argument" for God's existence, which uses the idea of necessary existence).
- **Computer Science**:
- **Temporal Logic** (a form of modal logic) is used in program verification and hardware design: one can assert things like "necessarily (if request then eventually grant)" meaning in all future execution paths, whenever a request happens, eventually a grant follows.
- **Modal logics of knowledge and belief** (epistemic logic) model what different agents know or believe in distributed systems or game theory.
- **Law and Ethics (Deontic Logic)**: There's a branch of modal logic for obligations and permissions. For example, $\Box p$ might mean "it is obligatory that p (p must be done)" and $\Diamond p$ meaning "it is permitted that p (p is allowed)". This is useful in formalizing rules and checking for conflicts (like ensuring something is not both obligatory and forbidden).
- **Planning and AI**: Reasoning about what could possibly happen vs what must happen can be captured in modal frameworks, especially in AI planning algorithms or exploring alternate scenarios.

A Taste of Modal Reasoning Structure

Modal logic often involves additional "axioms" depending on what kind of modality you're modeling. For instance: - **T axiom**: $\Box p \rightarrow p$ (if p is necessary, then p is true). Assumes our actual world is one of the possible worlds and follows the modal truth. This is usually assumed for most modalities like knowledge, physical necessity, etc. - **4 axiom**: $\Box p \rightarrow \Box \Box p$ (if p is necessary, then it's necessary that it's necessary). This can be intuitive for some notions (if something must be true, it must be unavoidably true).

that it must be true...). - **5 axiom:** $\Diamond p \rightarrow \Box \Diamond p$ (if p is possible, then it is necessarily possible — no new info will rule it out in other worlds).

These axioms define different *systems* of modal logic (S5, S4, etc.). For example, in an epistemic context, $\Box p$ might mean "I know p ". Then $\Box p \rightarrow p$ means "if I know p , then p is true" (a reasonable axiom: I can't know something that's false — at least in the normal definition of knowledge). Another axiom might be $\Box p \rightarrow \Box \Box p$ meaning "if I know p , then I know that I know p " (which might be debatable for human knowledge, but in idealized logic systems often assumed).

We won't go deep into these, but it's good to be aware that modal logic has different flavors depending on which axioms (constraints on the possible worlds relations) you assume.

Common Misconceptions

- **"Possibly p means p is true at least sometimes (in time)."**

Reality: "Possibly" in modal logic doesn't necessarily mean sometimes in time (that's temporal logic). It means in some conceivable scenario or under some assumption. So if I say "It's possible it's raining," I don't mean timewise, I mean there is a scenario consistent with what I know (or consistent with physical laws, etc.) where it's raining.

- **"Necessarily p means p is true no matter what, so p is just a tautology."**

Reality: $\Box p$ being true means given the modality in question (logical necessity, physical necessity, etc.), p holds in all those worlds. If we're talking logical necessity, yes then $\Box p$ means p is a tautology in propositional logic sense. But if we're talking physical necessity, $\Box p$ might not be a logical tautology, just a law-of-nature truth.

- **Mixing scopes:** People sometimes confuse $\Box(p \wedge q)$ with $\Box p \wedge \Box q$ (the latter implies the former, but the former doesn't necessarily imply the latter unless some frame conditions hold). Generally, $\Box$ distributes over $\wedge$ (since $\Box(p \wedge q) \rightarrow (\Box p \wedge \Box q)$ is valid), but $\Diamond$ does not distribute over $\vee$ similarly (actually $\Diamond(p \vee q) \equiv \Diamond p \vee \Diamond q$ is valid distribution for possibility over or). The logic rules can be tricky.

- **"Possible means we have no information."**

Reality: In some contexts, $\Diamond p$ can mean "for all we know, p could be true." But in strictly logical modal terms, it means "there is some accessible world (scenario) where p is true." It's not just ignorance, it's about whether a scenario exists or not under the modality's rules.

Exercises

10.1. Using modal operators, symbolize the statement: "It is possible that John will be late to the meeting." (Let L = "John is late to the meeting".)

10.2. Symbolize: "It is necessary that if the alarm is set then it will ring when triggered." (Let A = "the alarm is set", R = "the alarm rings when triggered". Hint: you want $\Box(\dots)$ of an implication.)

10.3. Consider the statement: $\Diamond(p \wedge q)$. In plain English, what does this mean? Provide a scenario (context) where such a statement might be used and make sense (you can make p and q represent simple conditions).

10.4. Does $\Diamond p \rightarrow p$ hold in general? If not, give a counterexample in plain terms (a situation where "possibly p " is true but p is false). What about the converse $p \rightarrow \Diamond p$?

10.5. If $\Box p$ is true, must $\Diamond p$ also be true? If $\Diamond p$ is true, must $\Box p$ be true? Explain in words.

10.6. Provide an example of a statement that is: - true, but not necessarily true (so p is true but $\neg \Box p$ is also true).
- false, but possibly true (so p is false but $\Diamond p$ is true).

(For instance, think of a factual statement about the world that is true but we can imagine a scenario where it's false, and vice versa.)

10.7. Translate into modal logic: "After studying the evidence, the detective concluded that the murderer must have entered through the window." (Hint: interpret "must have" as necessity relative to the detective's knowledge or the scenario. Let E = "the murderer entered through the window".)

10.8. Consider a simple game where either one player wins or the other. Express the rule "Necessarily, either Player1 wins or Player2 wins" in modal logic (let W_1 = "Player1 wins", W_2 = "Player2 wins"). Does this imply that it's not possible for the game

to tie? Explain.

10.9. (Challenge/Thinking) If $\Box p$ is true in our actual world, then p is true in all possible worlds (including this one). Does it follow that $\Box p \rightarrow p$ is a tautology of modal logic (always true no matter what p is)? Explain. (This relates to the axiom T in modal logic.)

10.10. Suppose we have $\Box(p \rightarrow q)$ and we also know $\Diamond p$. Can we infer $\Diamond q$? Explain intuitively and confirm with the idea of possible worlds. (Hint: if in every world $p \rightarrow q$ holds, and there's some world where p holds, in that same world q should hold, hence it's possible q .)

Congratulations on wading into the waters of modal logic! This is a more advanced topic, and we've only scratched the surface. Modal logic opens doors to many specialized systems (temporal, deontic, epistemic, etc.) each with their own interpretations of $\Box$ and $\Diamond$. But the fundamental idea is always to reason about what *must* be true versus what *might* be true.

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12: Logic Learning Resources

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Logic Learning Resource List

Welcome to the resource list for logic learning! Whether you're just beginning to explore formal logic or you're looking to deepen your understanding, the following curated resources will serve as your guide.

Disclaimer

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University Courses & Lectures

- **Stanford's Introduction to Logic** *Description:* This course provides a rigorous introduction to formal logic, exploring deductive reasoning, logical notation, and proof techniques. *Link:* [Stanford Intro to Logic](#)
 - **MIT OpenCourseWare: Mathematics for Computer Science** *Description:* Ideal for computer science students, this course covers discrete mathematics with a strong focus on logic, proofs, and combinatorics. *Link:* [MIT Mathematics for Computer Science](#)
 - **Harvard: Introduction to Logic** *Description:* A foundational course exploring symbolic logic, inference, and reasoning, designed for beginners as well as those looking to solidify their understanding. *Link:* [Harvard Logic Course](#)
 - **Coursera: Logic Courses** *Description:* Various universities offer logic courses on Coursera. These courses range from introductory to advanced levels and often include video lectures, quizzes, and peer discussions. *Link:* [Coursera Logic Courses](#)
-

Online Tutorials & Interactive Platforms

- **Khan Academy: Getting started with Logical Reasoning** *Description:* Free tutorials that break down the basics of logical reasoning. *Link:* [Khan Academy LSAT Logic](#)
 - **Brilliant.org: Logic** *Description:* Engaging, interactive lessons and puzzles designed to improve problem-solving and logical thinking skills. *Link:* [Brilliant Logic](#)
 - **YouTube Channels** *Description:* Explore lectures and explanatory videos from channels like **CrashCourse**, **Khan Academy**, and **MIT OpenCourseWare**. These channels offer visual and narrative explanations that make complex ideas accessible. *Search Tip:* Use keywords such as "Introduction to Logic" or "Logic for Beginners."
-

Books & Textbooks

- **"How to Prove It: A Structured Approach" by Daniel J. Velleman** *Description:* A popular textbook that teaches proof techniques and formal logical reasoning in an approachable manner. *Link:* [Amazon Link](#)
- **"A Concise Introduction to Logic" by Patrick J. Hurley** *Description:* A widely used text offering clear explanations of logical arguments and symbolic logic. *Link:* [Amazon Link](#)
- **"Logic in Computer Science: Modelling and Reasoning about Systems" by Michael Huth and Mark Ryan** *Description:* This book applies logical reasoning to computer science, covering topics like system modeling and verification. *Link:* [Amazon Link](#)
- **"An Introduction to Mathematical Logic" by Herbert Enderton** *Description:* A rigorous introduction to mathematical logic, ideal for those seeking a deeper theoretical perspective. *Link:* [Amazon Link](#)

- **"The Art of Reasoning" by David Kelley** *Description:* A guide to the fundamentals of logical thinking, suitable for both academic study and everyday reasoning. *Link:* [Amazon Link](#)
-

Online Communities & Reference Sites

- **Stanford Encyclopedia of Philosophy (Logic Section)** *Description:* An authoritative reference offering in-depth articles on various topics in logic and philosophy. *Link:* [Stanford Encyclopedia of Philosophy: Logic](#)
 - **ProofWiki** *Description:* A collaborative wiki that focuses on mathematical proofs and logical reasoning techniques. *Link:* [ProofWiki](#)
 - **Stack Exchange Communities** *Description:* Q&A platforms such as [Philosophy Stack Exchange](#) and [Mathematics Stack Exchange](#) provide community support for questions on logic.
 - **Reddit: r/logic and r/philosophy** *Description:* Online forums where enthusiasts and professionals discuss logic, debate ideas, and share resources. *Link:* [r/logic](#) | [r/philosophy](#)
-

Tools & Interactive Exercises

- **Truth Table Generators** *Description:* Use online tools to create and analyze truth tables, an essential step in understanding logical equivalence and validity. *Link:* [Truth Table Calculator](#)
 - **Logic Puzzle Apps** *Description:* Enhance your logical reasoning through engaging puzzles and games available on various app stores. *Search Tip:* Look for “logic puzzles” or “brain teasers” in your app marketplace.
 - **Interactive Proof Assistants** *Description:* Software like Coq or Lean helps users practice formal proofs and understand logical frameworks through interactive sessions. *Link:* [Lean Community](#)
-

Contact and Feedback

We value your input! Your feedback helps us refine and expand this resource list. Please use the embedded form below to share your thoughts, ask questions, or suggest additional resources.

Logic Learning Resources Feedback Form

Thank you for exploring Logic4All and our Logic Learning Resources! Your feedback is invaluable. Please fill out the following form to share your thoughts, ask questions, and suggest additional resources that could help others learn logic more effectively.

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Logic Glossary

Glossary Introduction

This glossary provides definitions for key terms used in logic, particularly first-order logic (FOL) and propositional logic. Understanding these terms is essential for learning formal logic systems and applying them to analyze arguments.

Using This Glossary

Terms are presented in alphabetical order with comprehensive definitions. Cross-references to related terms are included where applicable. This glossary covers terminology for both propositional and first-order logic, including syntax, semantics, proof methods, and logical properties.

Terms A-C

Antecedent: The antecedent of a conditional is its first component clause (the if clause). In $P \rightarrow Q$, P is the antecedent and Q is the consequent.

Antisymmetric: A binary relation R is antisymmetric iff no two things ever bear R to one another, i.e., R satisfies the condition that $\forall x \forall y [(R(x, y) \wedge R(y, x)) \rightarrow x = y]$.

Argument: "Argument" is used in two different senses in logic.

1. Arguments as pieces of reasoning: an argument is a sequence of statements in which one (the conclusion) is supposed to follow from or be supported by the others (the premises).
2. Arguments in the mathematical sense: an argument is an individual symbol (variable or constant) taken by a predicate in an atomic wff. In the atomic wff $\text{LeftOf}(x, a)$, x and a are

the arguments of the binary predicate LeftOf .

Aristotelian forms (A, E, I, O): The four main sentence forms treated in Aristotle's logic: the A form (universal affirmative) All P's are Q's, the E form (universal negative) No P's are Q's, the I form (particular affirmative) Some P's are Q's, and the O form (particular negative) Some P's are not Q's.

Arity: The arity of a predicate indicates the number of individual constants (names) it takes to combine with the predicate to form a complete sentence. A predicate with an arity of one is called unary. A predicate with an arity of two is called binary. It's possible for a predicate to have any arity, so we can talk about 6-ary or even 113-ary predicates.

Asymmetric: A binary relation R is asymmetric iff it is never reciprocated, i.e., R satisfies the condition that $\forall x \forall y (R(x, y) \rightarrow \neg R(y, x))$.

Atomic sentence: Atomic sentences are the most basic sentences of FOL, those formed by a predicate followed by the right number (see arity) of names. Atomic sentences of FOL correspond to the simplest sentences of English.

Biconditional: An if and only if statement. In FOL, the biconditional $P \leftrightarrow Q$ comes out true whenever P and Q have the same truth value and false when they differ in truth value. $P \leftrightarrow Q$ is equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$; that is, a biconditional is equivalent to a conjunction of "one-way" conditionals.

Biconditional Elimination ($\leftrightarrow$ Elim): A rule of systems F and FT that permits us, given a biconditional on one line and either of its components on another line, to infer the other component of the biconditional. Also known as modus ponens for the biconditional.

Biconditional Introduction ($\leftrightarrow$ Intro): A rule of systems F and FT that permits us to infer a biconditional $P \leftrightarrow Q$ from a pair of closed subproofs, one of which assumes P and deduces Q , the other of which assumes Q and deduces P .

Boolean connective: The logical connectives conjunction ($\wedge$), disjunction ($\vee$), and negation ($\neg$) allow us to form complex claims from simpler claims and are known as the Boolean connectives after the logician George Boole. Conjunction corresponds to the English word and, disjunction to or, and negation corresponds to the phrase it is not the case that. (See also Truth-functional connective.)

Bound variable: A bound occurrence of a variable is an instance of a variable occurring within the scope of a quantifier used with the same variable. For example, in $\forall x P(x, y)$ both occurrences of the variable x are bound, but y is free. (See also Free variable.)

Boxed constant: An individual constant placed inside a box when it is introduced in a subproof in system F. Boxed constants are used in conjunction with rules $\exists$ Elim and $\forall$ Intro to indicate that those constants denote arbitrary objects. The rules of system F do not permit a boxed constant to appear outside the subproof in which it is introduced.

Completeness: A formal system is complete if every valid inference is provable by means of the rules of the system. (See also Soundness.)

Complex noun phrase: A noun phrase containing more than just a single semantically significant word, such as noun + adjective, or adverb + adjective + noun. Examples of complex noun phrases would be small happy dog or politician who admires no Democrats. In FOL, we normally use truth-functional connectives and quantifiers in translating complex noun phrases.

Conclusion: The statement in an argument that is meant to follow from the other statements (the premises).

Conditional: An if ... then sentence, i.e., a sentence that expresses some kind of conditional relationship between the two parts of the sentence. Not all conditionals in a natural language, such as English, are

truth-functional. (See Material conditional, Truth-functional.)

Conditional Elimination ($\rightarrow$ Elim): A rule of systems F and FT that permits us to infer the consequent of a conditional sentence whose antecedent occurs alone on a separate line. Also known as modus ponens.

Conditional Introduction ($\rightarrow$ Intro): A rule of systems F and FT that permits us to infer a conditional sentence from the fact that we have proved the conditional's consequent in a closed subproof that has the conditional's antecedent as its assumption. (See Conditional proof.)

Conditional proof: A method of proof in which one proves a conditional sentence by assuming its antecedent and then deducing its consequent. (See Conditional Introduction.)

Conjunct: One of the component sentences in a conjunction. For example, A and B are the conjuncts of $A \wedge B$.

Conjunction: The Boolean connective $\wedge$ corresponding to the English word and. An FOL sentence whose main connective is $\wedge$ is also called a conjunction. Such a sentence is true if and only if each conjunct is true.

Conjunction Elimination ($\wedge$ Elim): A rule of systems F and FT that permits us to infer any of the conjuncts of a conjunction.

Conjunction Introduction ($\wedge$ Intro): A rule of systems F and FT that permits us to infer a conjunction from the fact that we have proved all of its conjuncts separately.

Consequent: The consequent of a conditional is its second component clause (the then clause). In $P \rightarrow Q$, Q is the consequent and P is the antecedent.

Contradiction Elimination ($\perp$ Elim): A rule of systems F and FT that permits us to infer any sentence we like from a contradiction.

Contradiction Introduction ($\perp$ Intro): A rule of systems F and FT that permits us to enter a contradiction, $\perp$, into a proof or subproof if we have already proved

both P and $\neg P$ on separate lines in that proof or subproof.

Contradiction symbol, $\perp$: The symbol $\perp$ represents contradiction, i.e., something that cannot possibly be true in any set of circumstances. An example would be the conjunction of a sentence and its negation, $S \wedge \neg S$. ($\perp$ can be pronounced simply "contradiction.")

Conversational implicature: An implicature of a speaker's assertion of a sentence S is a conclusion that a hearer might draw from the speaker's assertion of S, but that is not strictly part of the meaning of S.

Counterexample: An individual case or instance that falsifies a universal generalization. A counterexample to an argument is a possible situation in which the premises of the argument are true and the conclusion is false. Such a situation is therefore a counterexample to the generalization that the conclusion comes out true in all situations in which the premises come out true.

Terms D-F

Deductive vs. inductive: A deductive argument attempts to show that the conclusion is a logical consequence of the premises—that the conclusion must be true if the premises are true. An inductive argument does not attempt to show that the conclusion must be true, but only that its truth is made more probable by the truth of the premises.

Default and generous uses of rules (in Fitch): A default use of a rule is what Fitch does when you cite that rule and some previous line(s) as justification and, without entering any new sentence, ask Fitch to check out the step. For example, if you cite a conjunction and the rule $\wedge$ Elim and ask Fitch to check out the step, Fitch will enter the leftmost conjunct on the new line. A generous use of a rule is one that is not strictly in accordance with the rule as stated in F (i.e., F would not allow you to derive it in a single step), but is still a valid inference and will be approved of by Fitch.

Definite description: A phrase of the form the so-and-so that purports to refer to exactly one object, e.g., the king of Norway or the sum of 3 and 5.

DeMorgan's laws: There are two such laws. The first tells us that the negation of a conjunction, $\neg(P \wedge Q)$, is logically equivalent to the disjunction of the negations of the original conjuncts: $\neg P \vee \neg Q$. The second tells us that the negation of a disjunction, $\neg(P \vee Q)$, is logically equivalent to the conjunction of the negations of the original disjuncts: $\neg P \wedge \neg Q$.

DeMorgan laws for quantifiers: There are two such laws. The first tells us that the negation of a universal generalization, $\neg \forall x P(x)$, is logically equivalent to an existential generalization of a negation: $\exists x \neg P(x)$. The second tells us that the negation of an existential generalization, $\neg \exists x P(x)$, is logically equivalent to a universal generalization of a negation: $\forall x \neg P(x)$. These laws are also known as the quantifier/negation equivalences.

Disjunct: One of the component sentences in a disjunction. For example, A and B are the disjuncts of $A \vee B$.

Disjunction: The Boolean connective $\vee$ corresponding to the English word or. An FOL sentence whose main connective is $\vee$ is also called a disjunction. Such a sentence is true if and only if at least one disjunct is true.

Disjunction Elimination ($\vee$ Elim): A rule of systems F and FT that permits us to infer a sentence from a disjunction if we have inferred it from each disjunct. (See Proof by cases.)

Disjunction Introduction ($\vee$ Intro): A rule of systems F and FT that permits us to infer a disjunction from any of its disjuncts.

Distributing quantifiers: In some cases (but not all) an FOL sentence beginning with a quantifier is equivalent to the corresponding sentence with the quantifier "distributed through" the sentence. For example, given the sentence $\forall x (P(x) \wedge Q(x))$ we can distribute the universal quantifier through and obtain the equivalent sentence $\forall x P(x) \wedge \forall x$

$Q(x)$, and vice versa. This is called distributing $\forall$ through $\wedge$. One may also distribute $\exists$ through $\vee$. But it is not legitimate to distribute $\forall$ through $\vee$ or $\exists$ through $\wedge$.

Domain of discourse: The set of objects under consideration when claims involving quantifiers are evaluated—that is, the entire collection of things that we take the quantifiers to "range over" or pick out. For example, the truth value of the claim "Every student received a passing grade" depends on whether our domain of discourse includes all the students in the world, in the university, or just in one particular class. Also called Domain of quantification, or Universe of discourse. (See Infinite domain.)

Donkey sentence: An English sentence whose grammatical subject contains what appears to be an existential noun phrase (e.g., "a donkey") that serves as the grammatical antecedent to a pronoun (e.g., "it") in the verb phrase. These sentences get their name from the classic example every farmer who owns a donkey beats it. Donkey sentences are especially tricky to translate into FOL, and always require paraphrasing first. (See Paraphrasing.)

Common Examples

Classic examples of donkey sentences include:

- "Every farmer who owns a donkey beats it."
- "If a farmer owns a donkey, he beats it."

These sentences are challenging because the apparent existential quantifier "a donkey" behaves more like a universal quantifier in the logical structure.

Equivalence chain: A sequence of equivalent sentences designed to show the equivalence of the first and last sentences in the chain. In a typical equivalence chain, the equivalence between each sentence in the chain and the next is obvious, although the equivalence between the first sentence and the last is not.

Equivalence relation: An equivalence relation is a binary relation that is reflexive, symmetric, and transitive.

Exceptive: An exceptive is a claim that makes a universal generalization with an exception, such as everything is a cube except c. Exceptives go into FOL most conveniently as negative biconditionals, such as $\forall x (Cube(x) \leftrightarrow x \neq c)$.

Existential Elimination ($\exists$ Elim): A rule of system F that permits us to infer a sentence S from an existential generalization together with a closed subproof in which we have proved S from an instance of that generalization.

Existential Introduction ($\exists$ Intro): A rule of system F that permits us to infer an existential generalization from any of its instances.

Existential quantifier ($\exists$): In FOL, the existential quantifier is expressed by the symbol $\exists$ and is used to make claims asserting the existence of some object in the domain of discourse. In English, we express existentially quantified claims with the use of words like something, at least one thing, a, etc.

F vs. FT: F is the formal system of inference rules for FOL; FT is a part of F, and consists of just the rules for propositional logic. That is, FT is the set of introduction and elimination rules for $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, and $\perp$. F contains all of these rules plus the introduction and elimination rules for $\forall$, $\exists$, and $\equiv$.

First-order consequence: A sentence S is a first order consequence of some premises if S follows from the premises simply in virtue of the meanings of the truth-functional connectives, identity, and the quantifiers. More precisely, a sentence S is an FO consequence of sentences $P_1, \dots, P_n$ iff there is no interpretation under which all of $P_1, \dots, P_n$ come out true and S comes out false. (See Interpretation.)

First-order logic: A logical system in which quantifiers range over individuals, but not over properties or relations. A first-order logic thus contains individual variables, but not predicate variables.

First-order validity: A sentence S is a first order validity (FO validity, for short) if S is a logical truth simply in virtue of the meanings of the truth-functional connectives, identity, and the quantifiers. More precisely, a sentence S is an FO validity iff it comes out true on every interpretation. (See Interpretation, Validity.)

FO-contradiction: A sentence that comes out false simply in virtue of the meanings of the truth-functional connectives, identity, and the quantifiers. Every FO-contradiction is also a logical contradiction, but not conversely. (See also Logical contradiction, TT-contradiction, TW-contradiction.)

Formal proof: A step-by-step demonstration, given in a formal system of deduction, that a conclusion follows from its premises.

Free variable: A free occurrence of a variable is one that is not bound. (See Bound variable.)

Terms G-L

General Conditional Proof ($\forall$ Intro): A rule of system F that permits us to infer a generalized conditional sentence $\forall x (P(x) \rightarrow Q(x))$ from a closed subproof in which we have proved the instance of the consequent, $Q(c)$, from the assumption $P(c)$ for an arbitrary object c. (See Universal Introduction, Generalized conditional sentence.)

Generalized conditional sentence: An FOL sentence of the form $\forall x (P(x) \rightarrow Q(x))$, that is, a universal generalization of a conditional wff.

Inconsistent set of sentences: A set of sentences is called inconsistent if there is no possible circumstance in which they can all be true together. For an argument to be valid is for its premises together with the negation of its conclusion to form an inconsistent set of sentences.

Indirect proof: See Proof by contradiction.

Individual constant: The FOL version of a name—a symbol that stands for an object or individual. In FOL it is assumed

that each individual constant names one and only one object.

Infinite domain: A domain of discourse is infinite if it contains infinitely many members. The domain of positive integers is an example of an infinite domain. There are some invalid FOL arguments whose counter-examples must have infinite domains. Similarly, there are some FOL sentences that are logically possible but come out true only in infinite domains. (See Domain of discourse, Logically possible.)

Interpretation: If we take a quantified FOL sentence and replace its names with individual constants and its predicates with predicate letters, we get an uninterpreted FOL sentence. An interpretation of such a sentence is an assignment of values to those individual constants and predicate letters. Basically, an interpretation takes each individual constant to be a name of some object in the domain of discourse, and each predicate letter to stand for some property of objects in the domain. For example, from the FOL sentence $\forall x (\text{Cube}(x) \rightarrow \text{Larger}(x, b))$ we obtain the uninterpreted sentence $\forall x (F(x) \rightarrow R(x, b))$. One interpretation of this sentence takes the domain to be the positive integers, and assigns the number 2 to b , the property of being an even number to F , and the relation divisible by to R . So interpreted, the sentence says that every even number is divisible by 2.

Irreflexive: A binary relation R is irreflexive iff nothing stands in the relation R to itself, i.e., R satisfies the condition that $\forall x \neg R(x, x)$.

Literal: A literal is a sentence that is either an atomic sentence or the negation of an atomic sentence.

Logical consequence: A sentence S is a logical consequence of a set of premises if it is impossible for the premises all to be true while the conclusion S is false.

Logical contradiction: A sentence that comes out false in every possible circumstance. Every logical contradiction is also a TW-contradiction, but not

conversely. (See also TW-contradiction, FO-contradiction, TT-contradiction.)

Logical necessity: See Logical truth.

Logical truth: A sentence that is a logical consequence of any set of premises. That is, no matter what the premises may be, it is impossible for the conclusion to be false. A logical truth thus comes out true in every possible circumstance. This is also called logical necessity.

Logically equivalent sentences: Two sentences are logically equivalent if they have the same truth values in all possible circumstances.

Logically equivalent wffs: A pair of wffs with free variables are logically equivalent if, in any possible circumstance, they are satisfied by the same objects. (See Satisfaction.)

Logically possible: A sentence is logically possible if there is no logical reason it cannot be true, i.e., if there is a possible circumstance in which it is true.

Terms M-P

Material conditional: A truth-functional version of the conditional if ... then The material conditional $P \rightarrow Q$ is false if P is true and Q is false, but otherwise true.

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Metatheory: The branch of logic that studies the properties of a system of logic. Proofs of the soundness and completeness of system F , for example, belong to metatheory.

Mixed quantifiers: An FOL sentence that contains at least one universal quantifier and at least one existential quantifier is said to contain mixed quantifiers. When a sentence contains mixed quantifiers, the order in which the

quantifiers occur is especially important semantically.

Necessary and sufficient conditions: A necessary condition for a statement S is a condition that must hold in order for S to obtain. $S \rightarrow P$ says that P is a necessary condition for S . A sufficient condition for a statement S is a condition that guarantees that S will obtain. $P \rightarrow S$ says that P is a sufficient condition for S .

Negation: An FOL sentence that begins with a negation sign $\neg$ is called a negation. The negation of a true sentence is false; the negation of a false sentence is true.

Negation Elimination ($\neg$ Elim): A rule of systems F and FT that permits us to infer a sentence from the negation of its negation (e.g., to infer S from $\neg\neg S$).

Negation Introduction ($\neg$ Intro): A rule of systems F and FT that permits us to prove S by showing that $\neg S$ leads to a contradiction. (See Proof by contradiction.)

Null quantification: A null quantification is an FOL sentence beginning with a quantifier that does not bind any variables. That is, the variable used in the quantifier does not occur free in the wff to which the quantifier is prefixed. For example, the sentences $\forall x \text{Cube}(b)$ and $\forall y \exists y \text{Small}(y)$ are null quantifications. A null quantification is equivalent to the same sentence with the null quantifier deleted.

Numerical quantifier: English contains numerical quantifiers such as at least two, at most one, exactly one, at least three, at most two, exactly two, etc. FOL does not have numerical quantifiers, but we can paraphrase English sentences containing these phrases to enable their translation into FOL. These FOL translations use universal and existential quantifiers together with identity or nonidentity clauses to capture the meaning of the English numerical quantifiers.

Paraphrasing: To paraphrase a sentence is to say the same thing in a different way. It is often convenient to paraphrase an English sentence before attempting to translate it into FOL. For example, before

translating if something is a cube, it is not a tetrahedron into English, it is helpful to paraphrase it (e.g., as everything is such that, if it is a cube, it is not a tetrahedron) to make clear that the quantifier intended is universal.

Predicate symbol: A symbol of FOL that is used to express a property of objects or a relation between objects.

Prefix vs. infix notation: In prefix notation, the predicate or relation symbol precedes its arguments, e.g., $\text{Larger}(a, b)$. In infix notation, the relation symbol appears between its two arguments, e.g., $a = b$.

Premise: A statement meant to support (lead us to accept) the conclusion of an argument.

Prenex form: An FOL sentence is in prenex form when all its quantifiers are "out in front." More precisely, an FOL sentence is in prenex form when it contains no quantifier that is preceded by any symbol other than a quantifier. Thus, $\forall x \exists y (F(x) \rightarrow R(x, y))$ and $\text{Cube}(a)$ are both in prenex form. (The second example may seem surprising, but since it contains no quantifiers at all, it contains no quantifier that is preceded by any other symbol, and so counts as being in prenex form.) On the other hand, $\forall x (F(x) \rightarrow \exists y R(x, y))$ and $\forall x \neg \exists y R(x, y)$ are not in prenex form. Every FOL sentence is equivalent to some sentence in prenex form.

Presupposition: The presuppositions of a sentence S are those conditions that must be fulfilled in order for S to have a truth value, i.e., in order for S to make any claim at all. Whereas Russell's theory of descriptions holds that a sentence of the form F is G logically implies that there exists an F , Strawson's theory contends that such sentences presuppose the existence of an F , rather than logically imply it. So whereas Russell says that if nothing is F , the sentence the F is G is false, Strawson maintains that in this case the sentence is neither true nor false.

Proof by cases: A proof strategy that consists in proving some statement S from a disjunction by proving S from

each disjunct. (See Disjunction Elimination.)

Proof by contradiction (indirect proof):

To prove $\neg S$ by contradiction, we assume S and prove a contradiction. In other words, we assume the negation of what we wish to prove and show that this assumption leads to a contradiction. (See Negation Introduction.)

Proof by Contradiction Example

To prove "There is no largest prime number," we would:

1. Assume there IS a largest prime number (call it p)
2. Consider the number $n = (\text{product of all primes up to } p) + 1$
3. Show n is either prime or divisible by a prime larger than p
4. Either way, we reach a contradiction to our assumption

Therefore, there is no largest prime number.

Proofs without premises: A proof without premises, as the name implies, contains no premises. Such a proof typically begins with a subproof assumption and ends when all subproofs have been closed. The conclusion of a proof without premises is called a theorem of the system of proof. In a sound system, every theorem is a logical consequence of the empty set of premises, i.e., is a logical truth. (See Soundness, Theorem.)

Terms Q-S

Reflexive: A binary relation R is reflexive iff everything stands in the relation R to itself, i.e., R satisfies the condition that $\forall x R(x, x)$.

Satisfaction: An object named a satisfies an atomic wff $S(x)$ if and only if $S(a)$ is true, where $S(a)$ is the result of replacing all free occurrences of x in $S(x)$ with the name a . Satisfaction for wffs with more than one free variable is defined similarly. For example, an ordered pair of objects named a and b , respectively, satisfies an atomic wff $S(x, y)$ if and only if $S(a, b)$ is true, where $S(a, b)$ is the result of

replacing all free occurrences of x in $S(x, y)$ with the name a and all free occurrences of y with the name b .

Scope of quantifier: The scope of a quantifier in a wff is that part of the wff that falls under the "influence" of the quantifier. Parentheses play an important role in determining the scope of quantifiers. When a quantifier is immediately followed by a left parenthesis, the scope of that quantifier is the wff that is contained between that parenthesis and the right parenthesis that is its mate.

Semantics of quantified FOL: The semantics of a language concerns the meaning of its sentences and the conditions under which they come out true. For quantified FOL, the semantics consists of the rules for assigning truth values to sentences of FOL that are built up out of predicates, individual variables and constants, connectives, and quantifiers. The semantics of quantified FOL is based on the notion of satisfaction. (See Satisfaction.)

Sentence: In propositional logic, atomic sentences are formed by combining names and predicates. Compound sentences are formed by combining atomic sentences by means of the truth-functional connectives. In FOL, the definition is a bit more complicated. A sentence of FOL is a wff with no free variables.

Soundness: "Sound" is used in two different senses in logic.

1. An argument is sound if it is both valid and all of its premises are true.
2. A formal system is sound if all the inferences that are permitted by the rules of the system are valid inferences, that is, if no invalid arguments are provable within the system. (See also Completeness.)

Step-by-step method: A method of translating a natural language sentence into FOL that begins with the outer or "gross" structure of the sentence, and then moves inward, one step at a time.

Subproof: A proof that occurs within the context of a larger proof. In system F ,

subproofs are required by each of the following rules: $\forall$ Elim, $\neg$ Intro, $\rightarrow$ Intro, $\leftrightarrow$ Intro, $\forall$ Intro, and $\exists$ Elim.

Superlative: An adjective ending in *-est*, such as largest, oldest, strongest, etc. These are often translated into FOL in terms of the corresponding relational predicates (larger than, older than, stronger than), quantifiers, and identity.

Symmetric: A binary relation R is symmetric iff it is always reciprocated, i.e., R satisfies the condition that $\forall x \forall y (R(x, y) \rightarrow R(y, x))$.

Syntax of quantified FOL: The syntax of a language is, roughly speaking, its grammar—the rules for constructing wffs and sentences of the language. For quantified FOL, the syntax consists of the rules for constructing wffs using predicates, individual variables and constants, connectives, and quantifiers.

Terms T-W

T vs. $\vdash$: These "turnstile" symbols stand for the relation of provability in a formal system. $\vdash$ indicates provability in system F, while $\vdash$ indicates provability in system FT. We write $P_1, \dots, P_n \vdash T S$ to mean that there is a formal proof in FT of S from premises $P_1, \dots, P_n$, while we write $P_1, \dots, P_n \vdash S$ to mean that there is a formal proof in F of S from premises $P_1, \dots, P_n$. These symbols are sometimes written with no sentences to their left. When we write $\vdash T S$ we mean that S can be proved without premises in system FT, and $\vdash S$ means that S can be proved without premises in system F. (See F vs. FT.)

Tarski's World necessity: A sentence that comes out true in every world in Tarski's World. Such a sentence may not be logically necessary, since there might be a circumstance that cannot arise in Tarski's World in which it comes out false. An example would be $\neg(\text{Large}(b) \wedge \text{Larger}(a, b))$.

Tautological consequence: A sentence S is a tautological consequence of some premises if S follows from the premises simply in virtue of the meanings of the truth-functional connectives. We can

check for tautological consequence by means of truth tables, since S is a tautological consequence of the premises if and only if every row of their joint truth table that assigns true to each of the premises also assigns true to S . All tautological consequences are logical consequences, but not all logical consequences are tautological consequences.

Tautological equivalence: Two sentences are tautologically equivalent if they are equivalent simply in virtue of the meanings of the truth-functional connectives. We can check for tautological equivalence by means of truth tables since two sentences Q and S are tautologically equivalent if and only if every row of their joint truth table assigns the same value to the main connectives of Q and S .

Tautology: A sentence that is logically true in virtue of its truth-functional structure. This can be checked using truth tables since S is a tautology if and only if every row of the truth table for S assigns true to the main connective.

Theorem: In a formal system such as F or FT, a theorem is any sentence that has been proved within the system from the empty set of premises. (See also Proofs without premises.)

Total relation: A binary relation R is total iff for each object in the domain there is something to which it stands in the relation R , i.e., R satisfies the condition that $\forall x \exists y R(x, y)$.

Transitive: A binary relation R is transitive just in case if one thing bears R to a second, and the second bears R to a third, then the first thing bears R to the third, i.e., R satisfies the condition that $\forall x \forall y \forall z [(R(x, y) \wedge R(y, z)) \rightarrow R(x, z)]$.

Truth table: Truth tables show the way in which the truth value of a sentence built up using truth-functional connectives depends on the truth values of the sentence's components.

Truth-functional connective: A sentence connective with the property that the truth value of the newly formed sentence is determined solely by the truth value(s) of

the constituent sentence(s), nothing more. Examples are the Boolean connectives ($\neg$, $\wedge$, $\vee$) and the material conditional and biconditional ($\rightarrow$, $\leftrightarrow$).

Truth-functional form: The truth-functional form of a sentence is the structure that results from replacing all of its constituent quantified sentences with atomic sentence letters. Thus, the truth-functional form of $\forall x F(x) \rightarrow \forall x G(x)$ is $A \rightarrow B$, whereas the truth-functional form of $\forall x (F(x) \rightarrow G(x))$ is A .

TT-contradiction: A sentence that comes out false on every row of its truth table; that is, it comes out false simply in virtue of the meanings of the truth-functional connectives. Every TT-contradiction is also an FO-contradiction, but not conversely. (See also FO-contradiction, TW-contradiction.)

TT-possible ("truth table possible"): An FOL sentence is TT-possible if comes out true on at least one row of its truth table. Such a sentence may not be logically possible, but every sentence that is logically possible is also TT-possible.

TW-contradiction: A sentence that comes out false in every world in Tarski's World. Not every TW-contradiction is a logical contradiction. For example, $\text{Large}(b) \wedge \text{Larger}(a, b)$ is a TW-contradiction (since there is no Tarski world in which there is a block that is larger than a large block), but not a logical contradiction (since there are possible circumstances—not modeled by Tarski's World—in which one large object can be larger than another large object). (See also Logical contradiction, FO-contradiction, TT-contradiction.)

TW-possible ("Tarski's World possible"): A sentence that comes out true in at least one world in Tarski's World. Such a sentence is guaranteed to be both logically possible and truth table possible. But not all logically possible sentences are TW-possible (an example would be $\text{Large}(a) \wedge \text{Adjoins}(a, b)$, which is logically possible but not TW-possible).

Universal Elimination ($\forall$ Elim): A rule of system F that permits us to infer any

instance of a universal generalization.

Universal Introduction ($\forall$ Intro): A rule of system F that permits us to infer a universal generalization $\forall x P(x)$ from a closed subproof in which we have proved the instance $P(c)$ for an arbitrary object c . (See General Conditional Proof.)

Universal quantifier ($\forall$): In FOL, the universal quantifier is expressed by the symbol $\forall$ and is used to express universal claims. It corresponds, roughly, to English expressions such as everything, all things, each thing, etc.

Validity: "Validity" is used in two ways in logic.

1. Validity as a property of arguments:
An argument is valid if the conclusion

must be true in any circumstance in which the premises are true. (See also Logical consequence.)

2. Validity as a property of sentences: A first-order sentence is said to be valid if it is logically true simply in virtue of the meanings of its connective, quantifiers, and identity. (See First-order validity.)

Variable: Variables are expressions of FOL that function somewhat like pronouns in English. They are like individual constants in that they may be the arguments of predicates, but unlike constants, they can be bound by quantifiers. Generally letters from the end of the alphabet, x , y , z , etc., are used for variables.

Well formed formula (wff): Wffs are the "grammatical" expressions of FOL. A wff is either atomic (an n -ary predicate followed by n individual symbols) or a complex wff constructed using connectives, quantifiers, and other wffs. Wffs may have free variables. Sentences of FOL are wffs with no free variables. (See Bound variable.)

Working backwards strategy: A basic proof strategy in which you begin by figuring out what intermediate conclusion you might reach that would enable you to obtain your ultimate conclusion, and then take that intermediate conclusion as your new goal. You then repeat the strategy with the intermediate conclusion, and so on.

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